MPEG 1 H.261

MPEG1

- MPEG1 was an ISO standard (ISO/IEC 11172) developed to support VHS quality video at bitrate of ~1.5 Mbps. MPEG1 defined the syntax of encoding a stream video and the method for decoding.

- MPEG1 was developed for *progressive video* (non interlaced) so it manages only frames (progressive scan)
- Usual MPEG1 video resolution was: 352x240 or 320x240 at a bitrate of ~1.5 Mbps. This modality is referred to as *Constrained Parameters Bitstream* or CPB

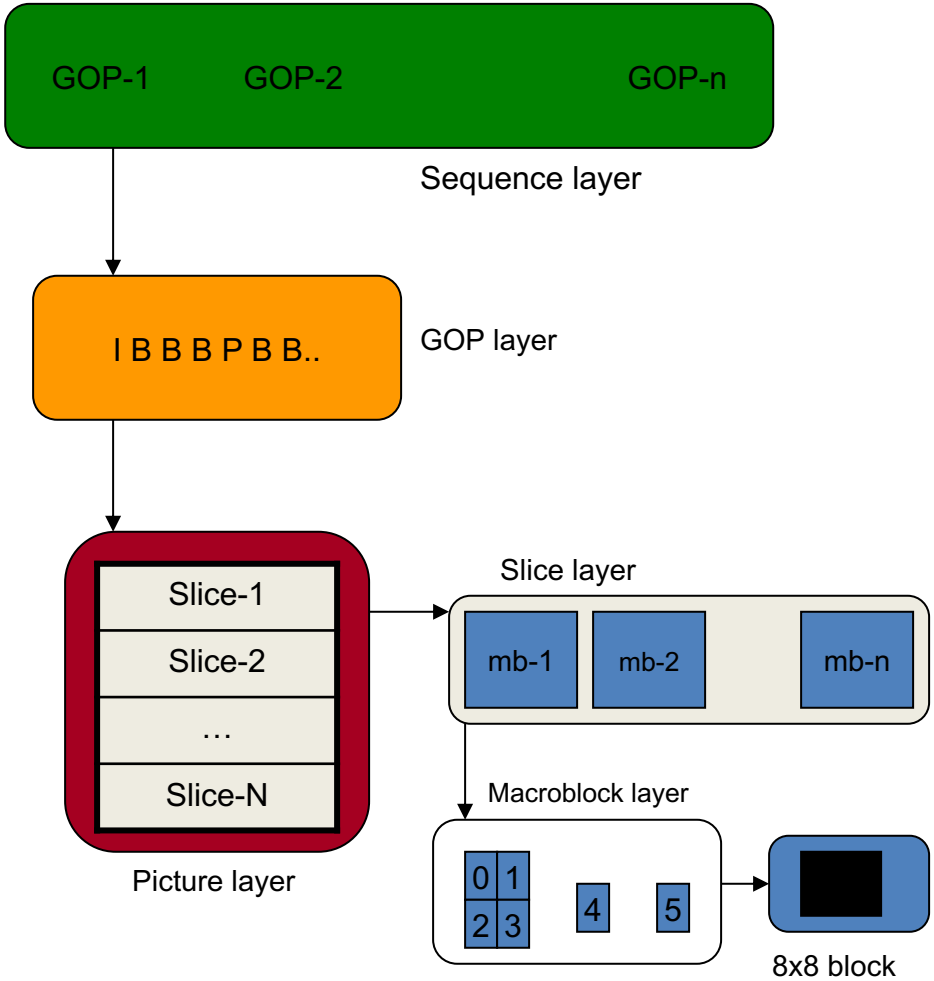
Resolution	Frames per Second
352×240	29.97
352×240	23.976
352×288	25
320×240^1	29.97
384×288^1	25

- MPEG-1 was based on the principle that an encoding of the differences between adjacent still pictures is a fruitful approach to compression. This is a general MPEG principle. It assumes that:
    - A moving picture is simply a succession of still pictures.
    - The differences between adjacent still pictures are generally small

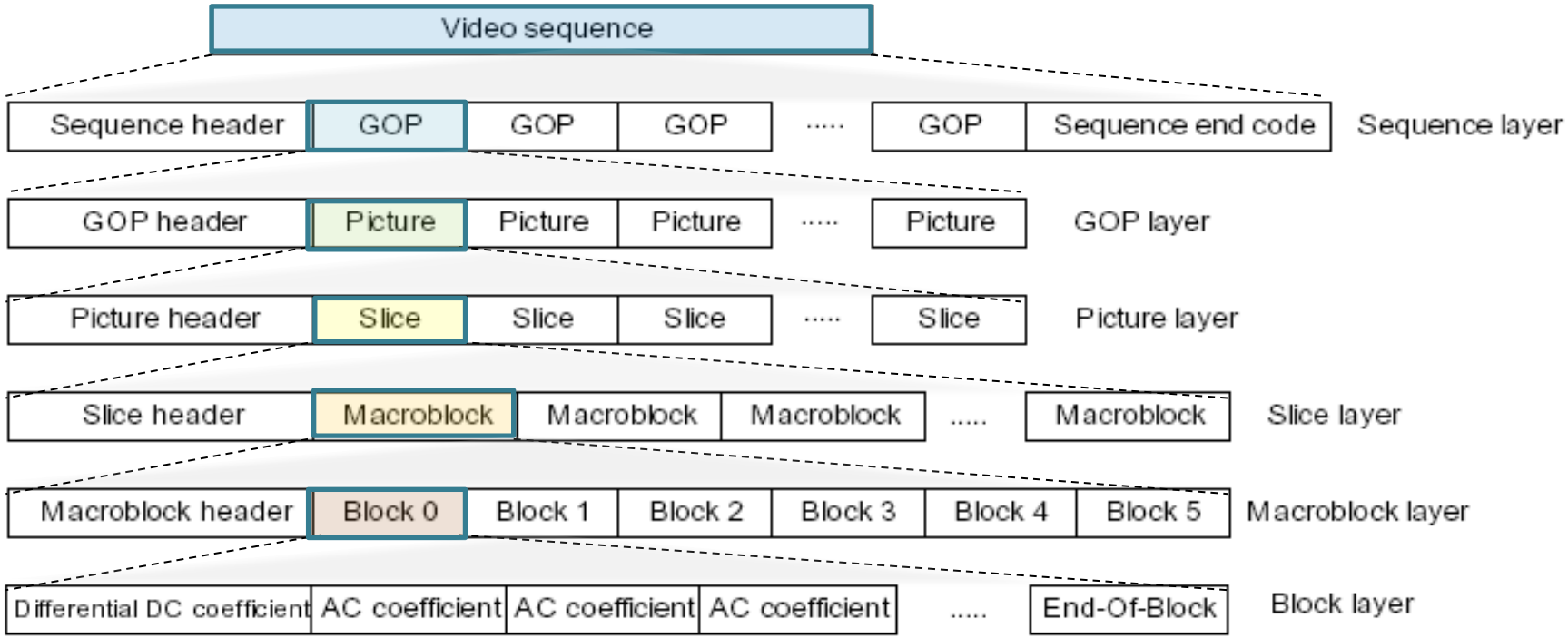
- Main features of MPEG-1:
    - *intra-frame coding*: for transform-domain-based compression. Similar to JPEG with 2D DCT, quantization and run-length encoding
    - *Inter-frame coding*: for block-based motion compensation. Similar blocks of pixels common to two or more successive frames are replaced by a pointer i.e. a *motion vector* that references one of the blocks. Predictive encoding is done with reference to an anchor frame according to interpolative techniques

MPEG1: Structure of the bit-stream

- *Sequence layer*: picture dimensions, pixel aspect ratio, picture rate, minimum buffer size, DCT quantization matrices
- *GOP layer*: unit for video random access; the smallest unit of independent coding. Contains encoded pictures (type I, P, B)
- *Picture layer*: primary coding unit. [temporal frame ref number, picture type, synchronization info, resolution, ….]
- *Slice layer*: syncronizzation unit. [position of slice in picture, quantization scale factor]
- *Macroblock layer*: motion compensation unit. [position of macroblock in slice, motion vectors (H, V)]
- *Block layer* : unit for DCT processing



MPEG-1 layers

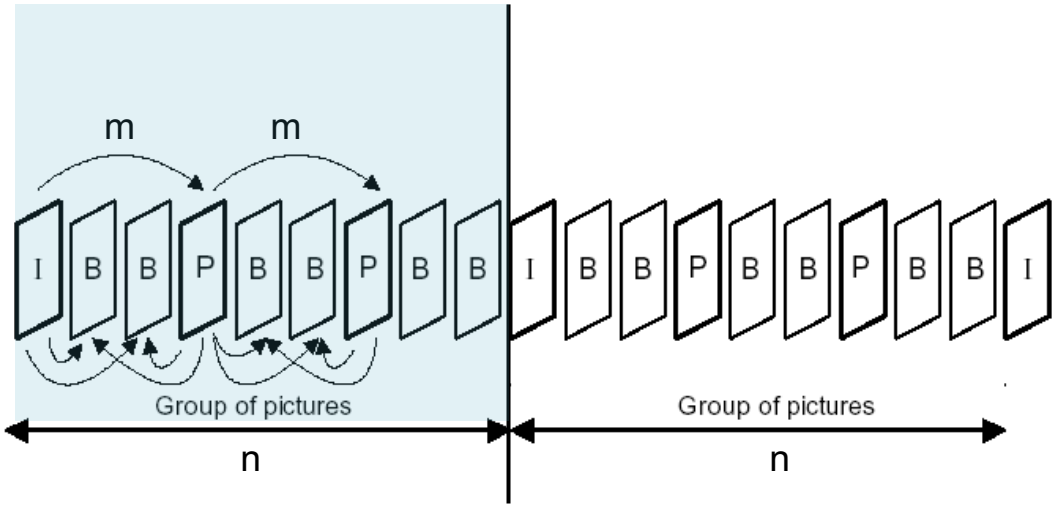


GOP

- A video sequence is decomposed in Groups of Pictures (GOPs). GOP is the distance between two keyframes, measured in the number of frames, or the amount of time between keyframes
  - Frames have different typology:
    - I (intra-coded), frames in full (also referred to as *keyframes*),
    - P (Predictive), B (Bi-directional) frames containing the difference between the keyframe and the subsequent frames
    - D (DC) contain DC coefficients only and are used for preview exclusively
  - Distance between I, P e B frames can be defined when coding.

- A GOP is *closed* if can be decoded without information from frames of the preceding GOP. Max GOP lenght are 0.5-2 secs.

Typically m=3, n=9:

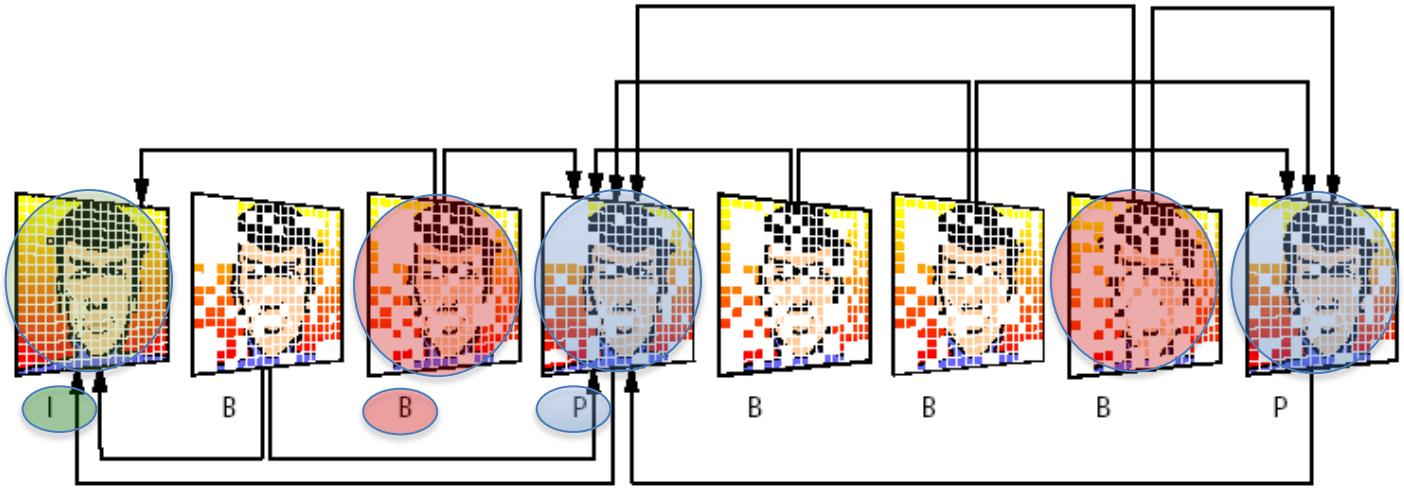


Impact GOP settings on video quality

Frame of the same video encoded at 2.5Mbps with a GOP of 4 frames (top) and a GOP of 90 frames (bottom)

Because there are fewer B and P frames in the top example, the encoder has to compress them more to fit within the configured bitrate, which results in blockiness, blurriness, and loss of detail.

Pictures



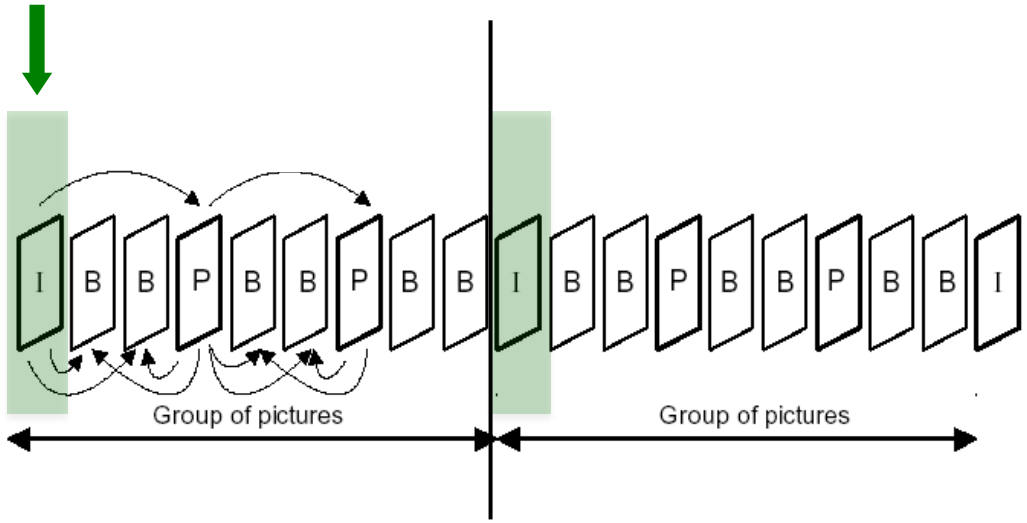
- I-frame: contains the full image
- P-frame is based on preceding I or P-frame and contains encoding of residual error after prediction
- B-frame uses past and future I or P frames and contains encoding of residual error after prediction

I-frames

- *Intra–coded* (I-)*frames* are so called because they are decoded independently from any other frames (they contain the full image). They are identical to JPEG frames.

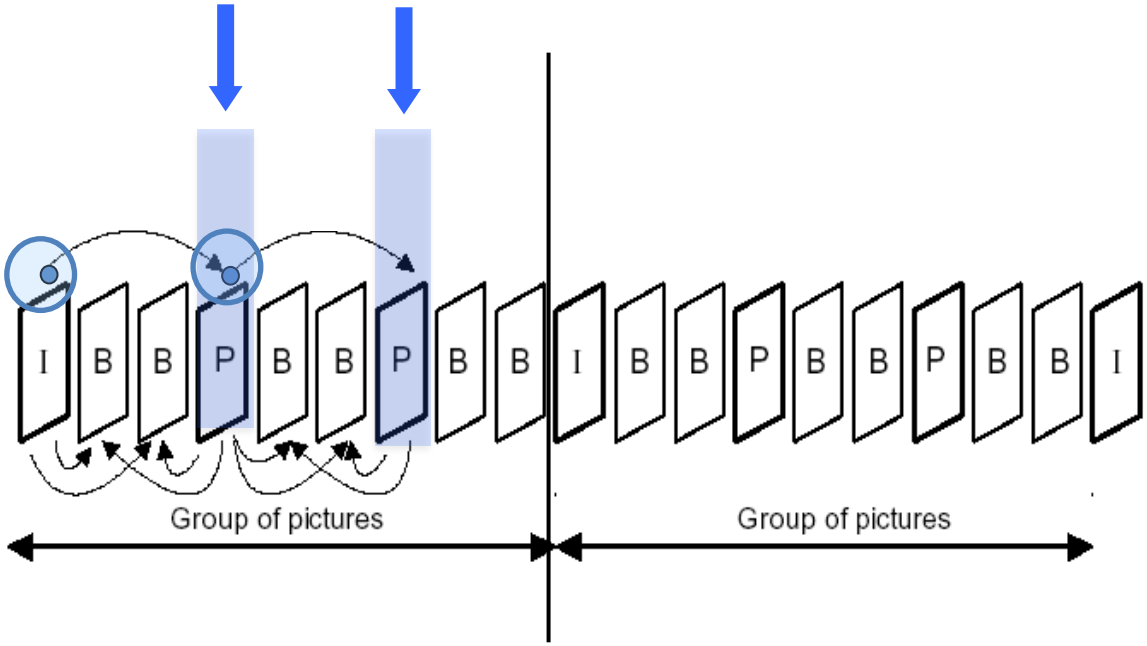
- Intra-Coded frame are coded with no reference to other frames (*anchor*).
  Minimize propagation of errors and permit random access. I-frame compression is very fast but produces large files (three times larger than normally encoded MPEG video)

- D frames: are similar to I frames but are only DC encoded (no AC coefficients). They are low quality representations used as thumbnails in video summaries



P-frames

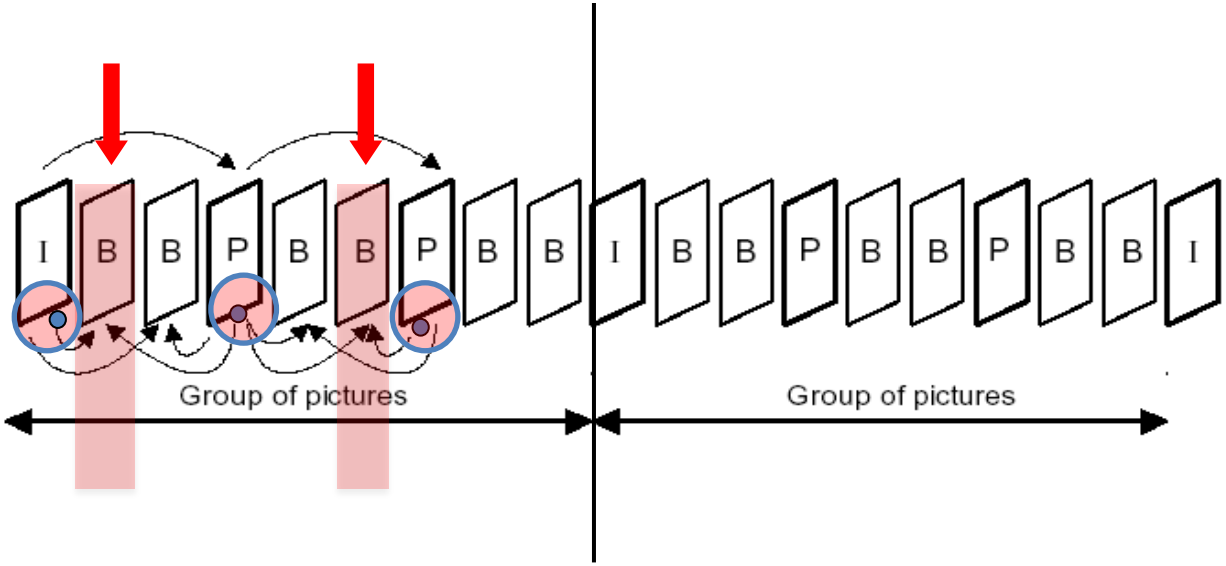
- *Predictive-Coded* (P-)*frames* are coded with forward motion prediction from preceding I o P frame.
- Improve compression by exploiting the temporal redundancy. They store the difference in image from the frame immeditely preceding it. The difference is calculated using *motion vectors*.



B-frames

- *Bi-directional-Coded* (B-)*frame* are coded with bidirectional (past and future) motion compensation using I and P frame (no B frame).

- Motion is inferred by averaging past and future predictions. Harder to encode introduces delay in coding. The player must first decode the next I or P frame sequentially after the B frame before it can be decoded and displayed. This makes B frames computationally complex and requires large data buffers.



Picture type and compression

- Relative number of (I), (P), and (B) pictures can be arbitrary. It depends on the nature of the application. It may depend on fast access and compression ratio requirements:
  - relatively smaller amount of compression is expected to be achieved at (I) pictures compared to (P) and (B) pictures.
  - the (B) pictures are expected to provide relatively the largest amount of compression under favorable predict

Considerations about picture types

- *Random access:* Sending the first frame as a keyframe and subsequent differences could work if every user started from the first frame and only watched forward. More keyframes need to be placed throughout the video to allow viewers to begin watching from random access points.

- *Error resiliency:* sending only differences from what came before, errors continue to propagate through the rest of the video if an error or corruption occurs. Adding additional keyframes provides error resiliency by returning the decoder to a known good frame and clearing previous errors that may have been propagating.

- *Scene changes:* Sending only differences between frames works very well when the differences between frames are relatively small. During content changes, it's possible for nearly the entire image to be filled with new information so making no sense to continue sending only differences. A video encoder will often detect this situation and automatically insert a new keyframe at the boundary point.

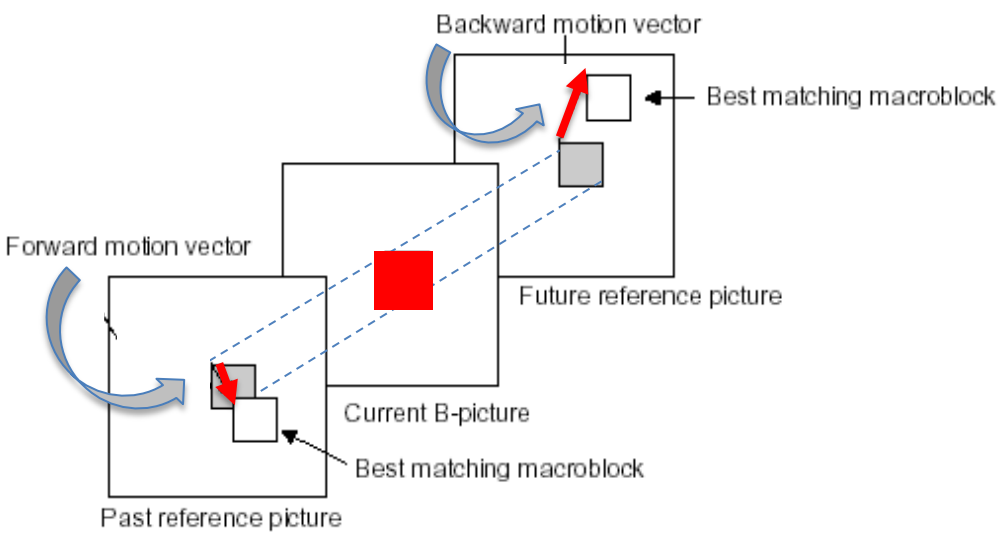
Frames and macroblocks

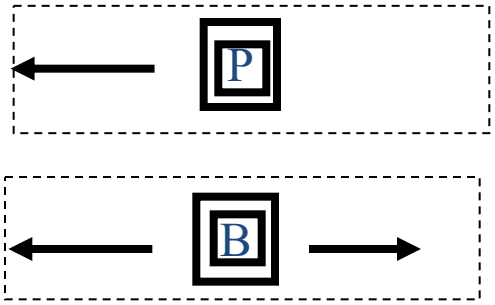
- Each video *frame* contains *macroblocks* that is the smallest independent unit of video considered by MPEG.

- Macroblocks are set of 16x16 pixel and are necessary for purposes of the calculation of motion vectors and error blocks for motion compensation.

Macroblocks

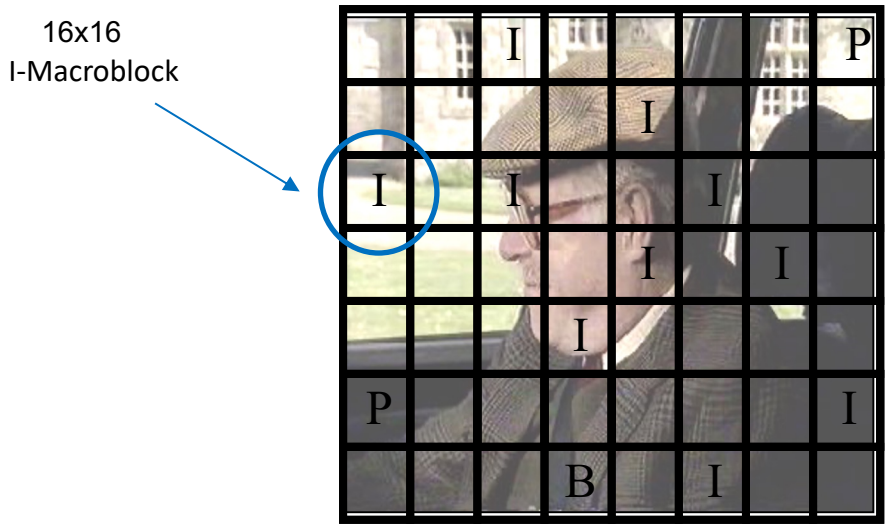
- Main types of macroblocks:
  - I *macroblocks*: encoded independently of other macroblocks (by 2D Discrete Cosine Transform as in JPEG blocks)
  - P *macroblocks*: encode not the region but the motion vector and error block of the previous frame (*forward predicted macroblock*)
  - B *macroblocks*: same as above except that the motion vector and error block are encoded from both the previous (*forward predicted macroblock*) and next frame (*backward predicted macroblock*)





Frames and macroblocks

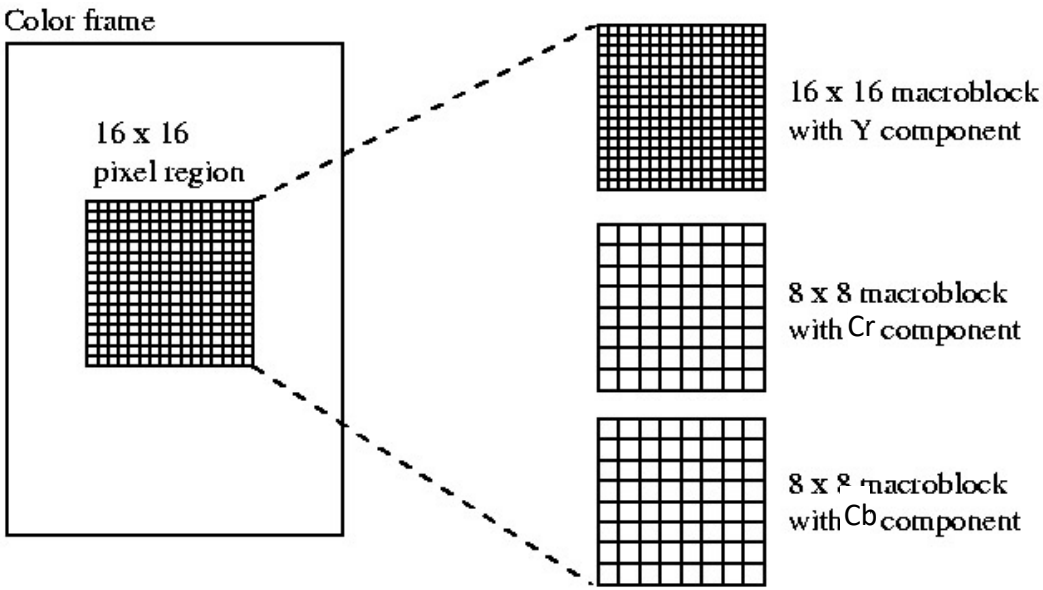
- Different types of macroblocks apply to different type of frames:
  - I and D frames: contain the full image
    contain I-macroblocks only with direct encoding from the image samples
  - P and B frames: contain encoding of residual error after prediction:
    - P frames: contain I or P macroblocks
    - B frames: contain I, P or B macroblocks



B Frame with I, P, B macroblocks

Macroblock components

- Each macroblock is encoded separately.

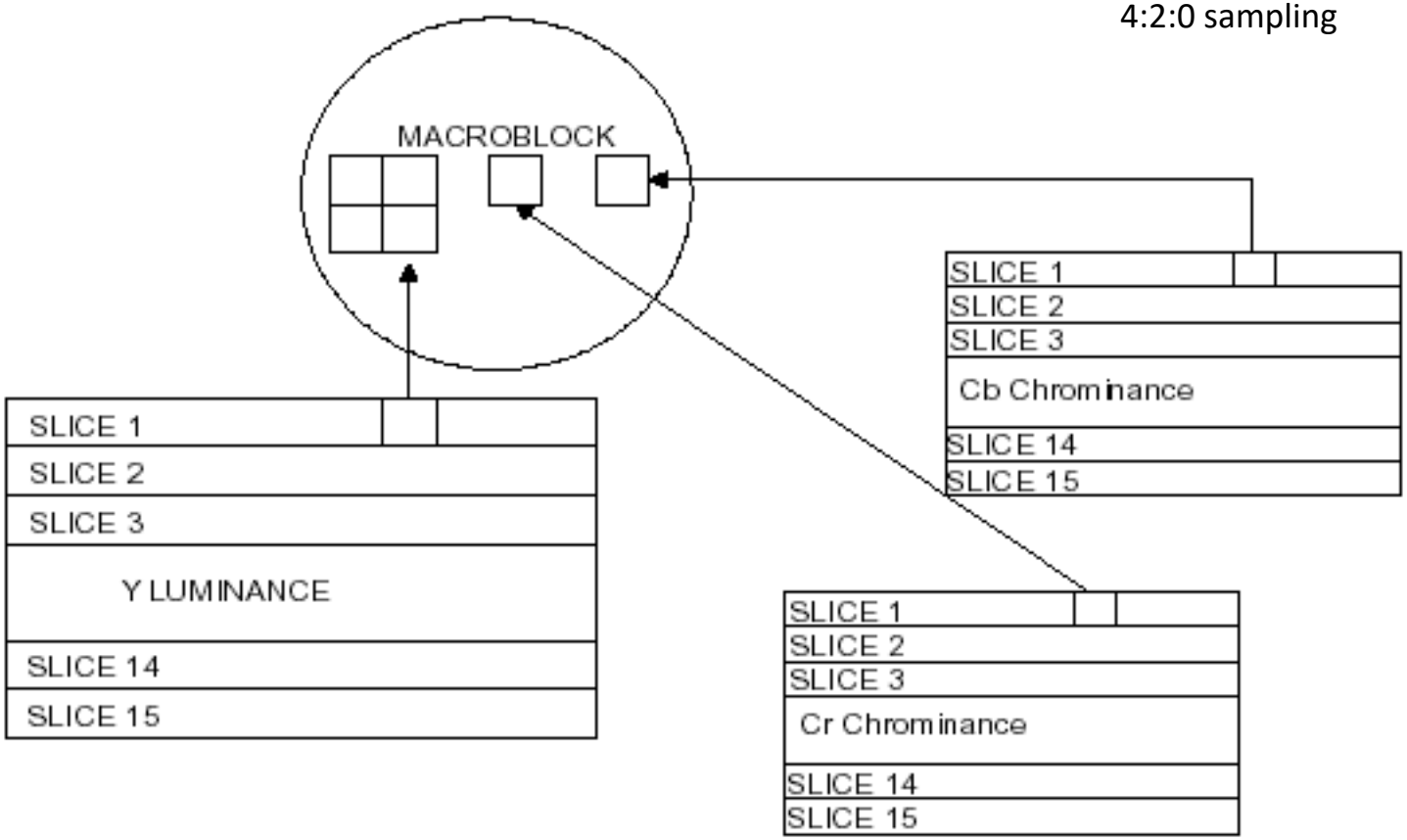


4:2:0 sampling

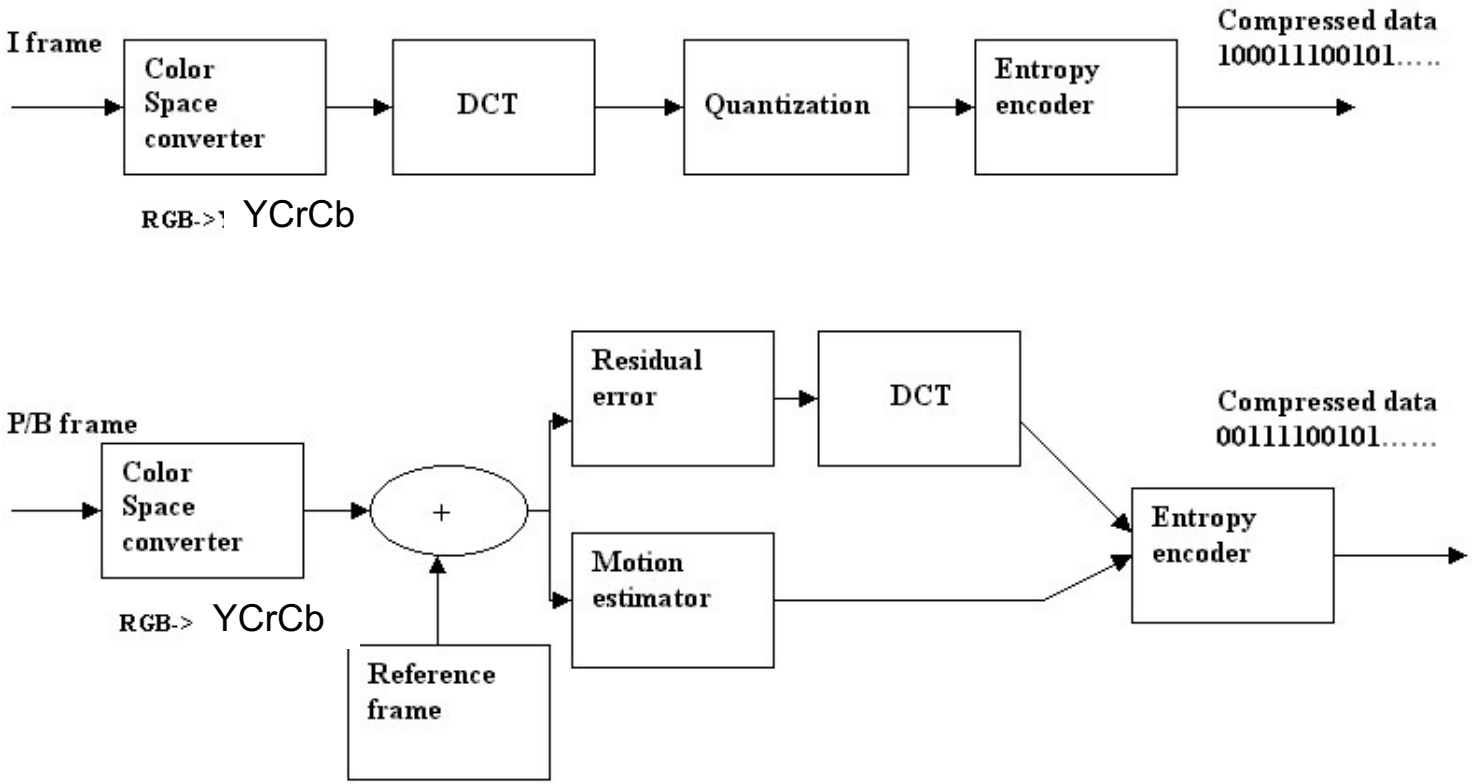
Slices

- Macroblocks are organized into slices

4:2:0 sampling

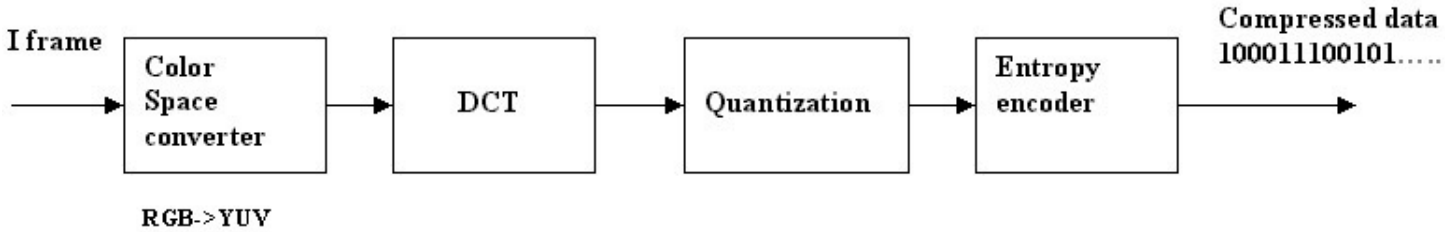


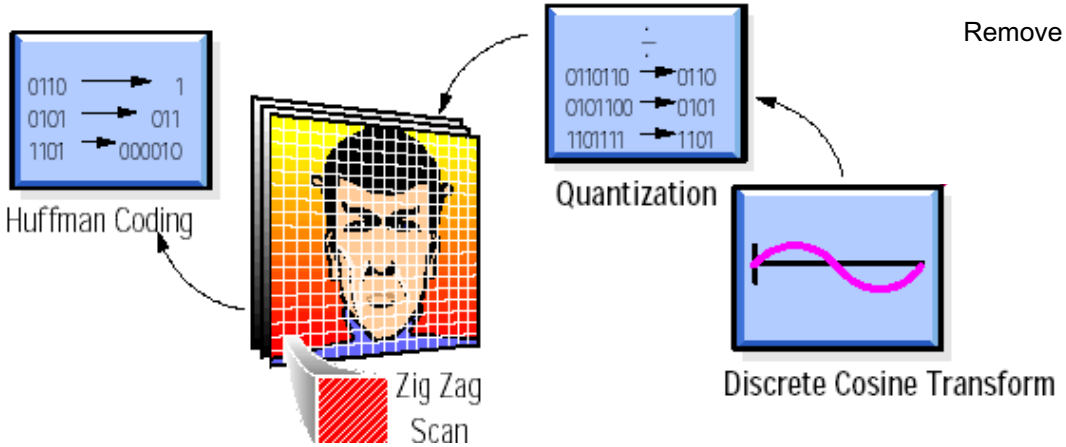
Encoding macroblocks



The block diagram of the MPEG encoder

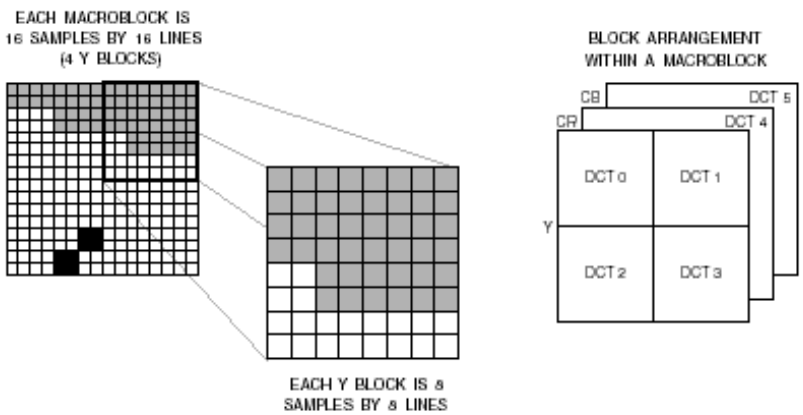
I macroblock coding



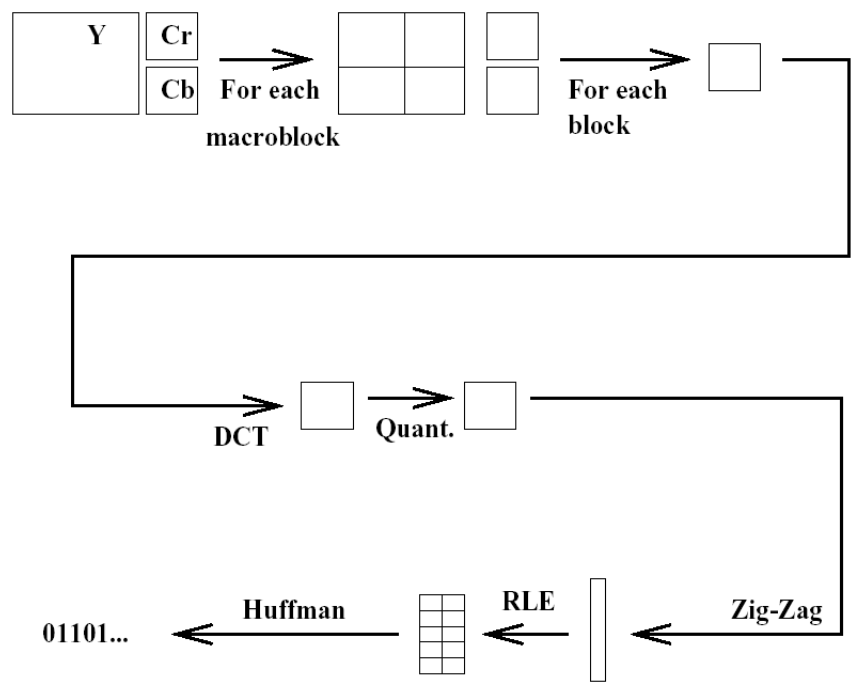


Remove spatial redundancy

I-macroblock coding



4:2:0 sampling



- Intra blocks are processed through DCT 8x8
- DCT coefficient quantization (lossy)
- zig-zag scanning
- DC (DPCM) and AC (RLE) coding
- Entropy coding (Huffman)

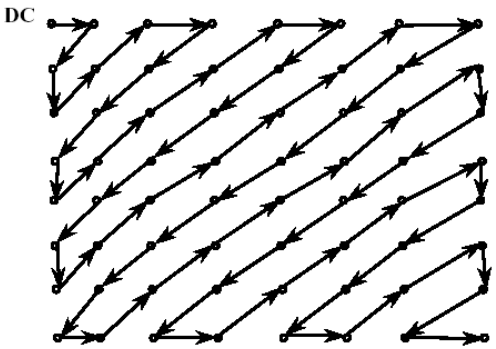
Spatially-Adaptive Quantization

- Spatially-adaptive quantization is an option made possible by the *factor quantizer_scale*. This parameter is allowed to vary from one macroblock to another within a picture to adaptively adjust the quantization on a macroblock basis based on variance.

8	16	19	22	26	27	29	34
16	16	22	24	27	29	34	37
19	22	26	27	29	34	34	38
22	22	26	27	29	34	37	40
22	26	27	29	32	35	40	48
26	27	29	32	35	40	48	58
26	27	29	34	38	46	56	69
27	29	35	38	46	56	69	83

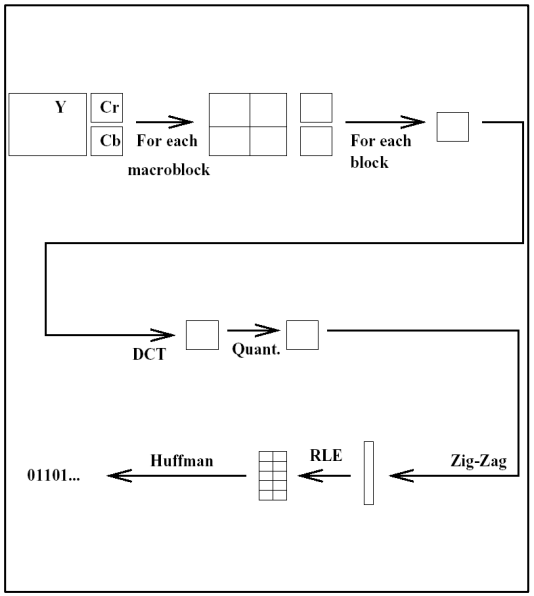
MPEG1 default quantization matrix

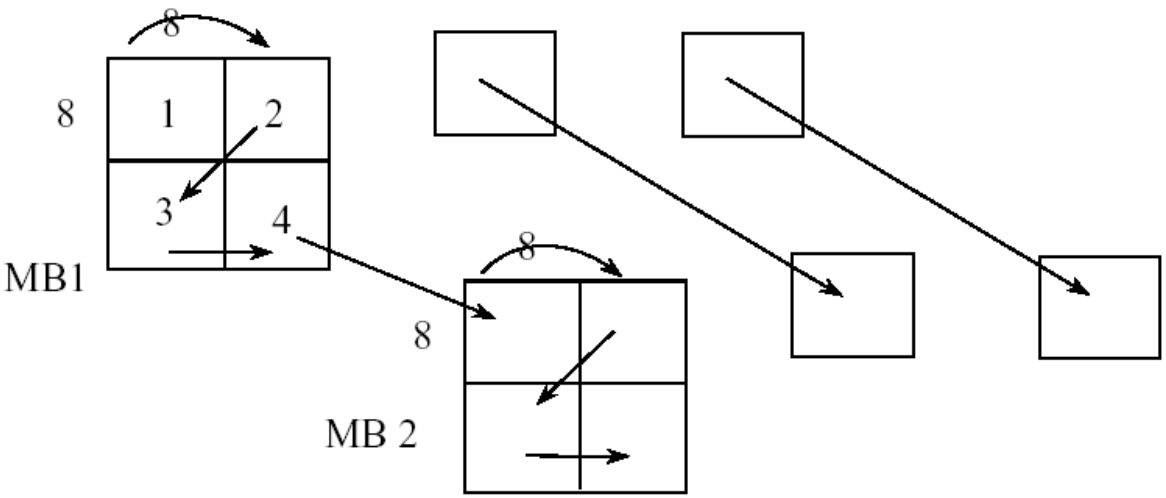
- zig-zag scanning is used to create a 1D stream



Coefficient encoding

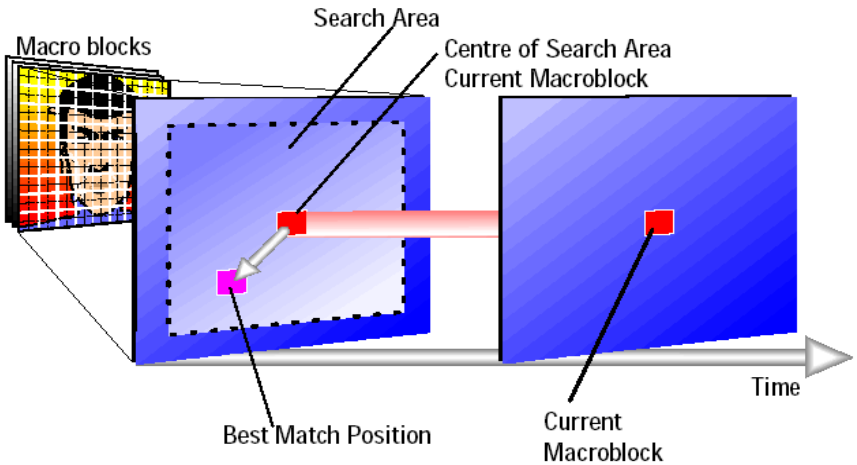
- AC coefficients are encoded losslessly according to *run length encoding* and *Huffman coding* with different tables for Y and CbCr.
- DC coefficients encode differences between blocks of the macroblock as in figure

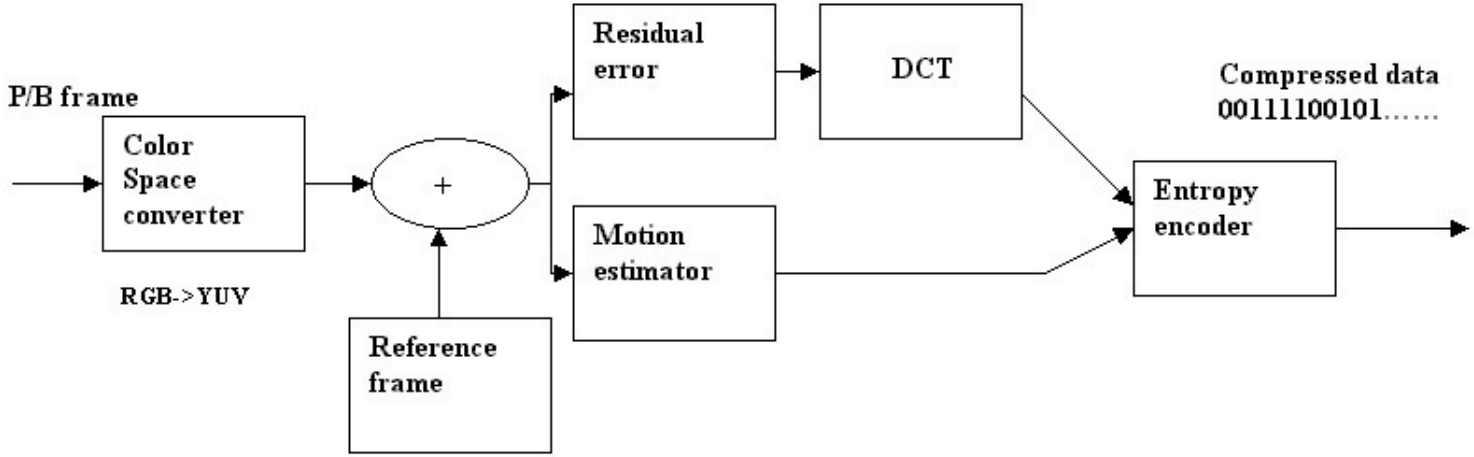




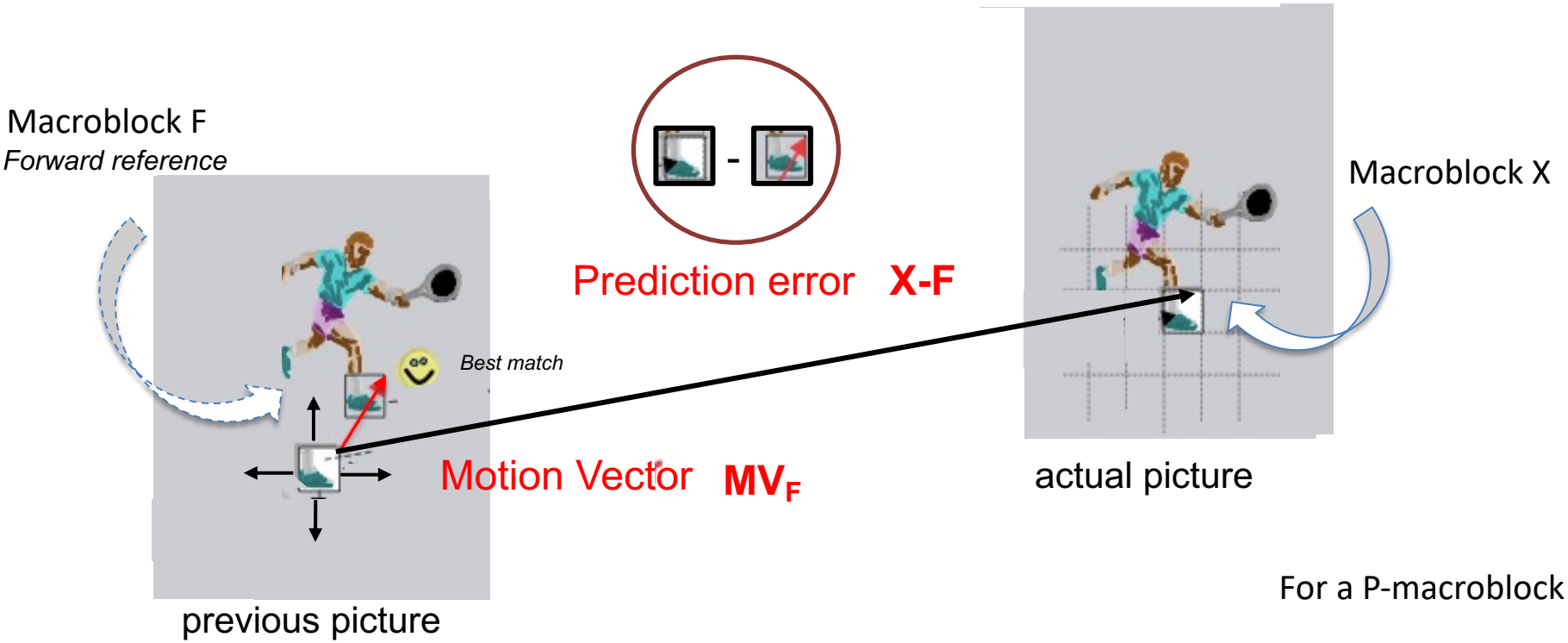
P/B macroblock coding

Remove temporal redundancy





Predictive encoding



- Predictive encoding aims to reduce the data transmitted by detecting the motion of objects. This will typically result in 50% - 80% savings in bits.

- Instead of sending quantized DCT coefficients of macroblock X:
  - Find the best-matching block in the reference frame by searching an area and compare. Each block can be assigned a match from either a *backward* (B) or *forward* (F) *reference*
  - Send quantized DCT coefficients of prediction error X-F: if prediction is good, error will be near zero and will need few bits.
  - Encode and send the motion vector MV_F.

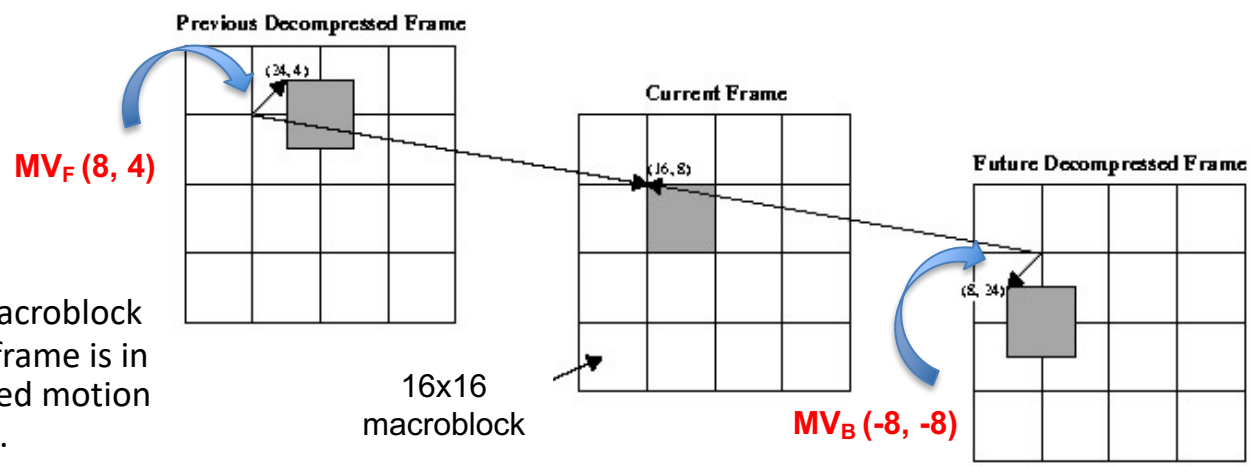
Block motion compensation

- The process of replacing blocks with a motion vector and the error block is referred to as *block motion compensation*. P and B macroblock coding is based on block motion compensation
- The component of a macroblock for motion compensation is the luminance Y component.
  - A *motion vector* describes the transformation between the same (similar) blocks in adjacent frames in a video sequence. Motion vectors are assumed to be *constant over a macroblock*
  - The encoder must decide whether a macroblock is encoded as I o P. Appropriate algorithms are used f.e. comparing the variance of luminance of the original macroblock with the *error macroblock*.

Motion vectors

- A motion vector is specified with two components (*horizontal* and *vertical* offset). Absence of motion vector is indicated with (0,0).
  - Offset is calculated starting from the top left pixel :
    - Positive values indicate top or right move.
    - Negative values indicate bottom or left move.
  - Set to 0,0 at the start of the frame or slice or I-type macroblock.

- Motion vectors are reset when a new I macroblock is found.

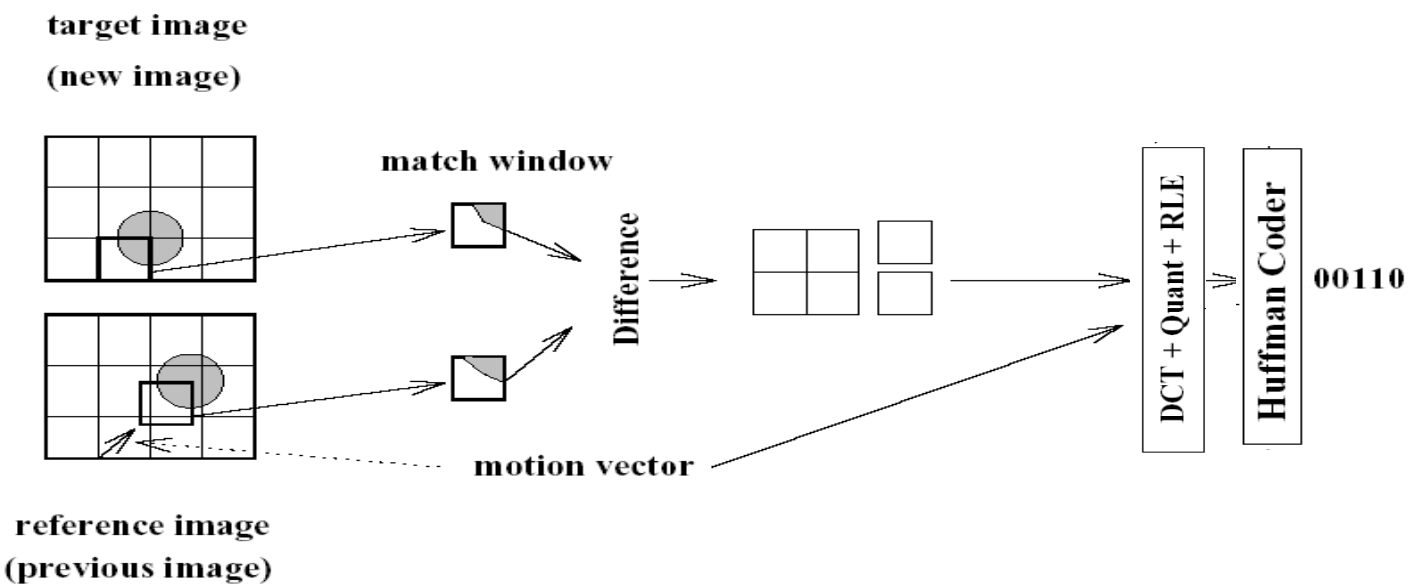


Example: the match of the shaded macroblock of the current frame in the previous frame is in position (24,4). The forward predicted motion vector for the current frame is (8, 4)…

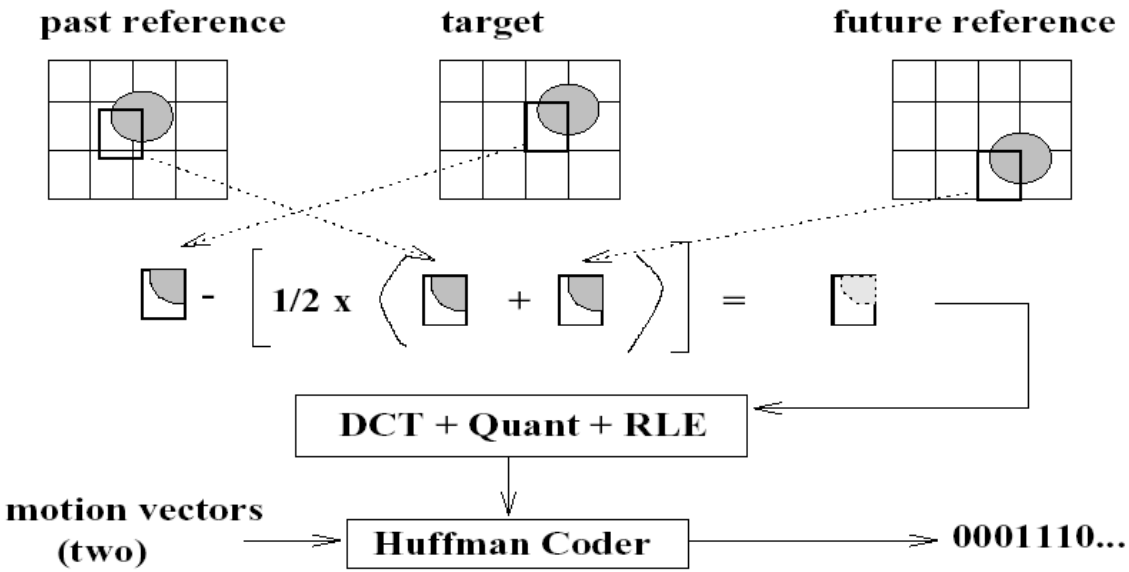
Error blocks

- The error block is obtained as the difference between two motion compensated blocks in adjacent frames. It is encoded as a normal block.
- For P/B error blocks a different quantization matrix is used wrt I-blocks: “16” value is set in all the matrix positions as error blocks have usually high frequency information
- DC component and AC component are managed in the same way (there is no differential encoding as in I blocks)

- For a P macroblock:



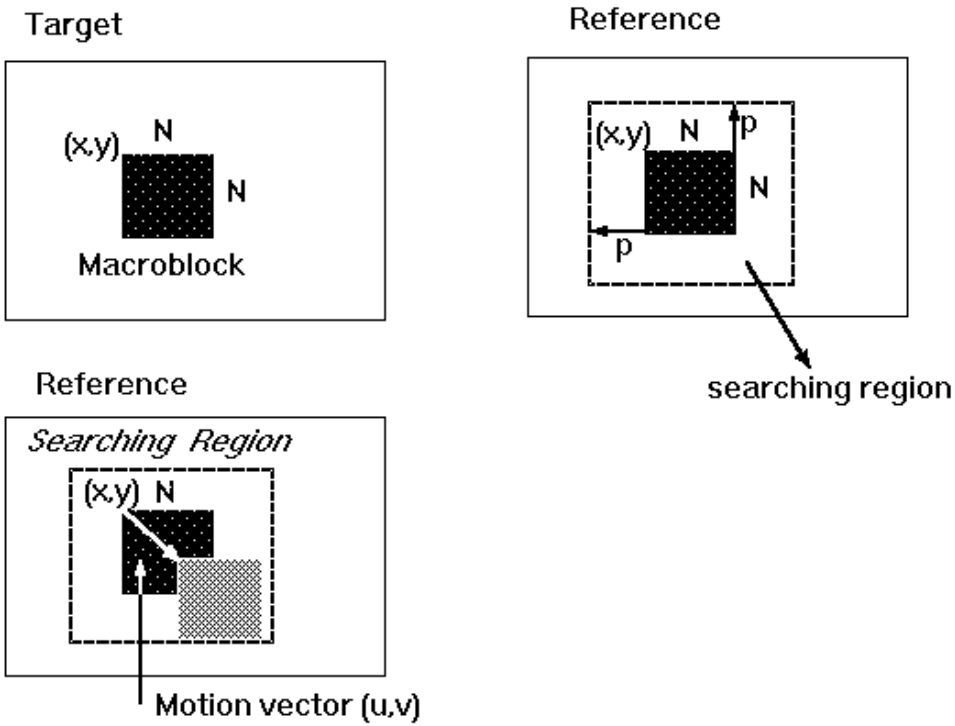
- For a B macroblock:



Motion estimation by block matching

- Motion estimation is performed by applying block matching algorithms.

- Matching is assessed according to different criteria.
- Different block matching techniques exist: often they limit the search area for matching.



MSE and SSD block matching criterion

- *Mean Squared Error* (MSE):

$$MSE = 1/N^2 \sum_{ij} (C_{ij} - R_{ij})^2$$

- *Sum of Squared Differences* (SSD):

$$SSD = \sum_{ij} (C_{ij} - R_{ij})^2$$

C_{ij} the sample in the current block
R_{ij} the sample in the reference block

MSE example

1	3	2
6	4	3
5	4	3

Current block

1	3	2	4	5
6	4	2	3	2
5	4	2	2	3
4	4	3	3	1
4	6	7	4	5

Search area

$$\begin{bmatrix} (-1,1) & (0,1) & (1,1) \\ (-1,0) & (0,0) & (1,0) \\ (-1,-1) & (0,-1) & (1,-1) \end{bmatrix}$$

Relative positions wrt (0,0)

$$MSE = 1/N^2 \sum_{ij} (C_{ij} - R_{ij})^2$$

$$\{(1-4)^2 + (3-2)^2 + (2-3)^2 + (6-4)^2 + (4-2)^2 + (3-2)^2 + (5-4)^2 + (4-3)^2 + (3-3)^2\}/9 = 2.44$$

Block centered in:

MSE value:

Position (x, y)	$(-1, -1)$	$(0, -1)$	$(1, -1)$	$(-1, 0)$	$(0, 0)$	$(1, 0)$	$(-1, 1)$	$(0, 1)$	$(1, 1)$
MSE	4.67	2.89	2.78	3.22	2.44	3.33	0.22	2.56	5.33

minimum value

MAE and MPC block matching criterion

- *Mean Absolute Error* (MAE):
  Easier than MSE

$$MAE = 1/N^2 \sum_{ij} | C_{ij} - R_{ij} |$$

- *Sum Absolute Differences* (SAD)
  Easier than SSD

$$SAD = \sum_{ij} | C_{ij} - R_{ij} |$$

C_{ij} the sample in the current block
R_{ij} the sample in the reference block

- *Matching Pel Count* (MPC):

Similar pixels are counted in the two blocks

SSD vs SAD

- Sum of Squared Differences (SSD) is more sensitive to outliers wrt Sum of Absolute Differences (SAD):

$$SSD = \sum_{ij} (C_{ij} - R_{ij})^2$$

$$\begin{matrix} 7 & 9 & 8 \\ 5 & 4 & 6 \\ 9 & 8 & 2 \end{matrix} \quad \text{versus} \quad \begin{matrix} 8 & 7 & 10 \\ 6 & 5 & 4 \\ 10 & 7 & 1 \end{matrix} \quad \text{-> SSD = 18}$$

$$\begin{matrix} 7 & 9 & 8 \\ 5 & 4 & 6 \\ 9 & 8 & 2 \end{matrix} \quad \text{versus} \quad \begin{matrix} 8 & 7 & 10 \\ 6 & 5 & 4 \\ 10 & 7 & \boxed{202} \end{matrix} \quad \text{-> SSD = 40,017}$$

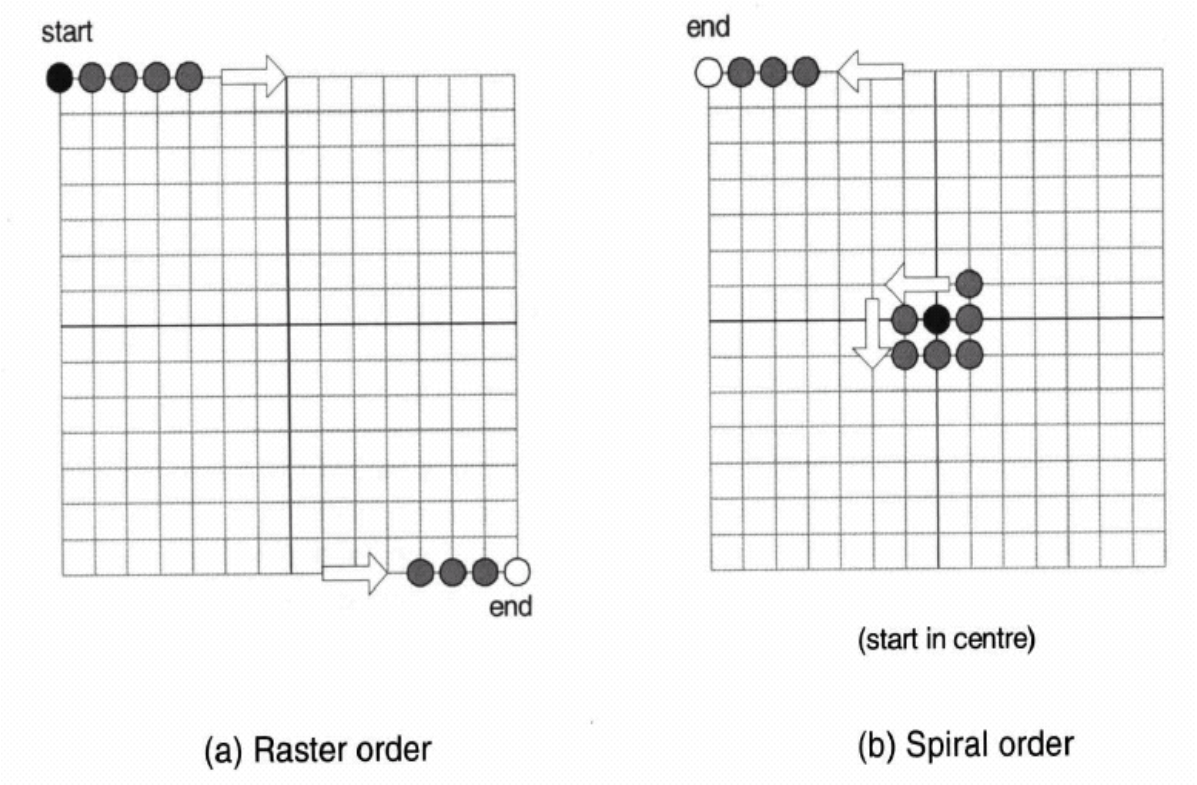
Outlier

$$SAD = \sum_{ij} | C_{ij} - R_{ij} |$$

$$\begin{matrix} 7 & 9 & 8 \\ 5 & 4 & 6 \\ 9 & 8 & 2 \end{matrix} \quad \text{versus} \quad \begin{matrix} 8 & 7 & 10 \\ 6 & 5 & 4 \\ 10 & 7 & 202 \end{matrix} \quad \text{-> SAD = 211}$$

Full search matching technique

- Full search technique always detects the global minimum
- All the positions within the window are checked with a pre-defined criterion for block matching. Computationally expensive, only suited for hardware implementation



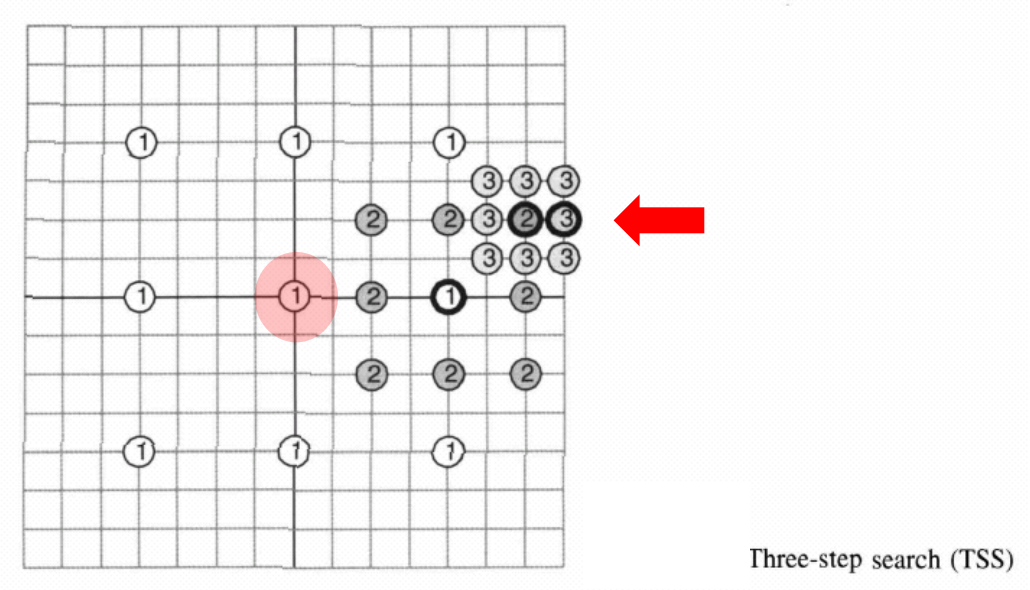
Fast search matching techniques

- As a less expensive alternative, Fast search methods employ a reduced number of comparisons wrt full search but may fall into local minima:
  - Three step search (TSS)
  - Logarithmic Search
  - One-at-a-Time Search
  - Nearest Neighbours Search

Three step search

1. Start search from (0, 0).
2. Set $S = 2^{N-1}$ (step size).
3. Look within 8 locations at +/-S pixel distance around (0, 0).
4. Select minimum SAD location between the 9 that have been analyzed
5. This location is the center for the new search
6. Set *S* = S/2.
7. Repeat from 3 to 5 until *S* = 1.

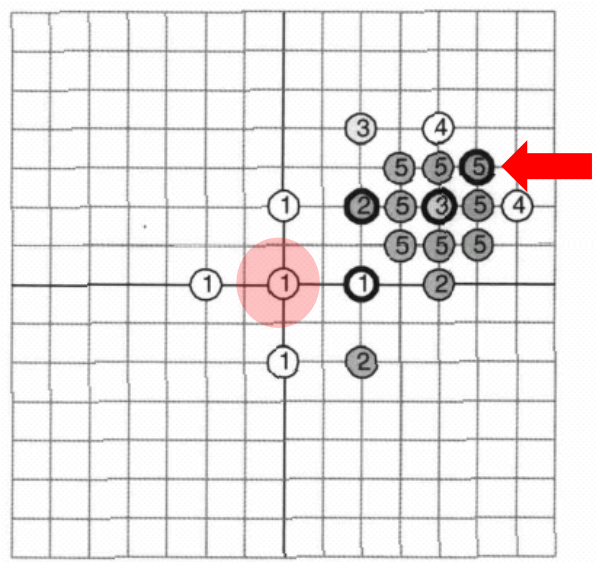
Step size = 4



Logarithmic Search

1. Start search from (0, 0).
2. Search in the 4 adjacent positions in the horizontal and vertical directions, at S pixel distance from (0,0) (S search step). The 5 positions model a ‘ + ’.
3. Set the new origin at the best match. If best match is in the central position of ‘+’ then *S* = S/2, otherwise *S* is not changed.
4. If *S* = 1 go to *5 , otherwise go to* 2.
5. Look for the 8 positions around the best match. Final result is the best match between the 8 positions and the central position

Step size = 2

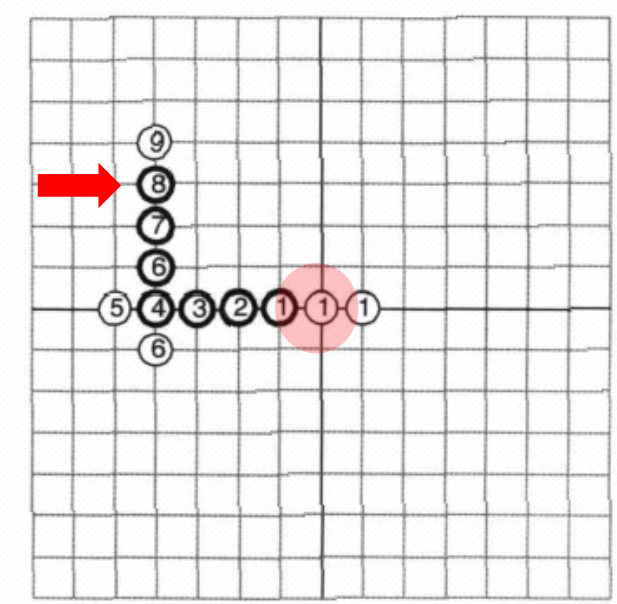


Logarithmic search

One-at-a-Time Search

1. Start from (0, 0).
2. Search at the origin and in the nearest positions horizontally
3. If origin has the lowest SAD then go to 5, otherwise. . . .
4. Set origin at the lowest SAD horizontally and search in the nearest position not yet checked and go to 3.
5. Repeat from 2 to 4 vertically.

Step size = 1

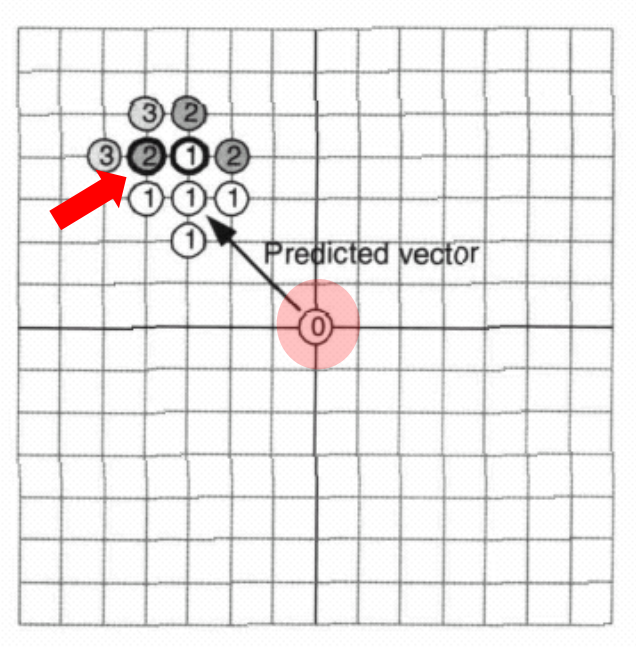


One-at-a-time search

Nearest Neighbours Search

- Motion vectors are predicted by the near vectors already coded (used in MPEG-4). Assumes that near macroblocks have similar motion vectors

1. Start from (0, 0).
2. Set origin in the position of the predicted vector and start from there
3. Search in the nearest '+ '.
4. If the origin is the best then take this position as the correct one. Otherwise take the best match and proceed
5. Stop when the best match is at the center of ' + ' or at the border of the window.



Block matching algorithms comparison

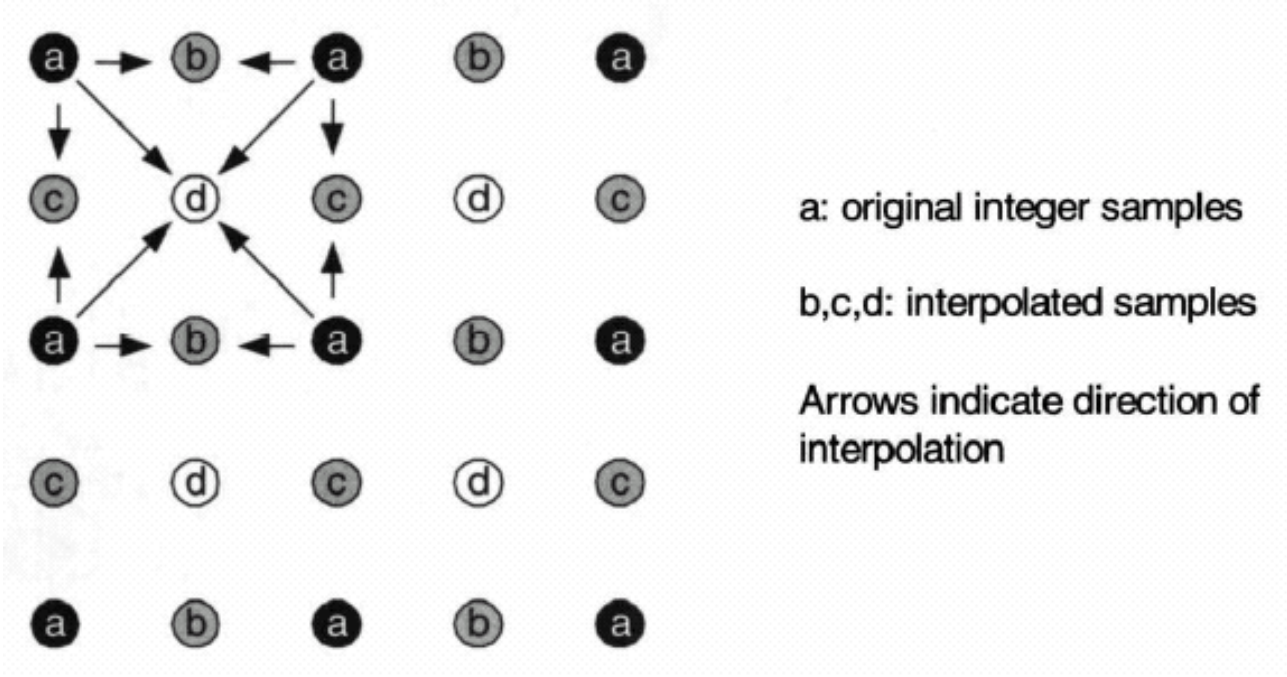
- Logarithmic search, cross-search e one-at-a-time have low computational complexity and low matching performance as well.

- Nearest-neighbours search, has good performance, similar to full search, and moderate computational complexity

Sub pixel motion estimation

- In some cases matching is improved if search is performed in an artificially generated region that is obtained by interpolating the pixels of the original region. In this case accuracy is sub-pixel.

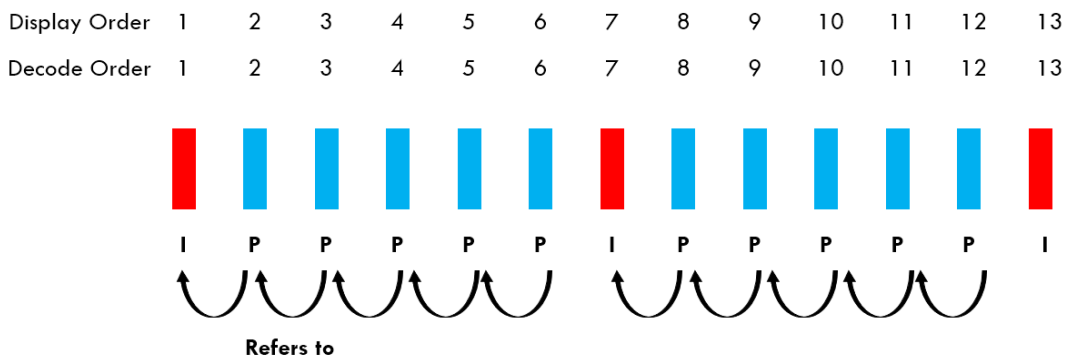
- Searching is performed as follows:
    1. Pixels are interpolated in the image search area so that a region is created with higher resolution than the original.
    2. Best match search is performed using both pixel and subpixel locations in the interpolated region
    3. Samples of the best matched region (full- o sub-pixel) are subtracted from the samples of the current block to obtain the error block.



Half pixel interpolation

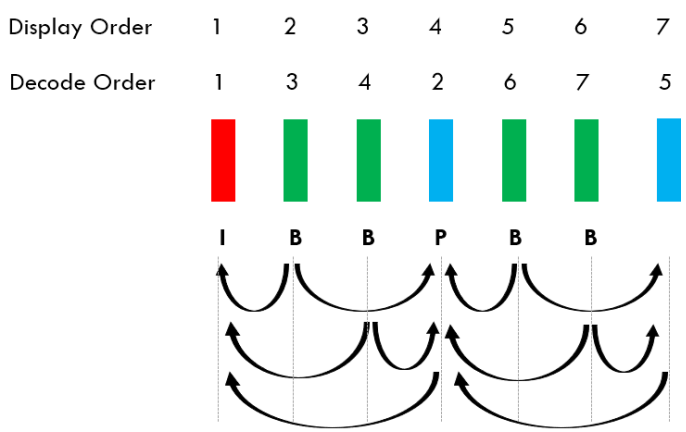
MPEG1 decoding and display

- In MPEG pictures are coded and decoded in a different order than they are displayed: this is due to bidirectional prediction for B pictures. The encoder needs to reorder pictures because B-frames always arrive late.



With P-frames only frames can come in and out of the encoder in the same order.

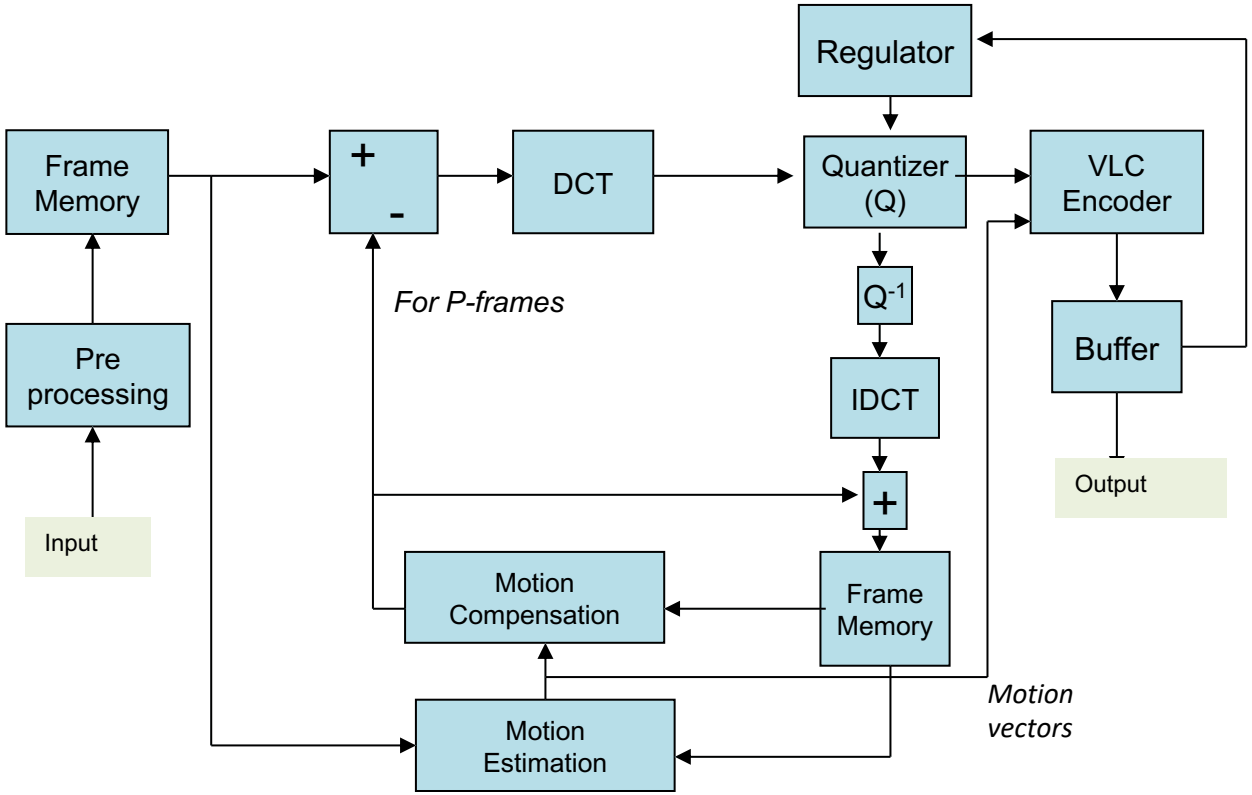
The encode/decode and display order is the same



With P- and B-frames a B-frame refers to another frame that is displayed in the future, we need to wait till that frame enters the encoder, and is encoded.

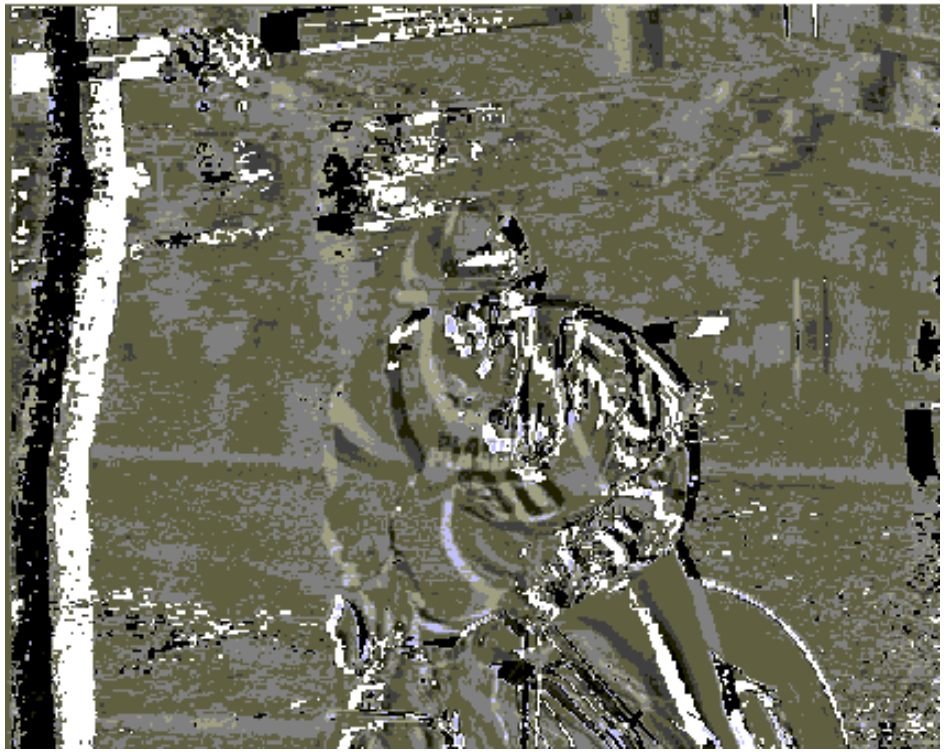
The encode/decode and display orders are different

The MPEG1 encoder

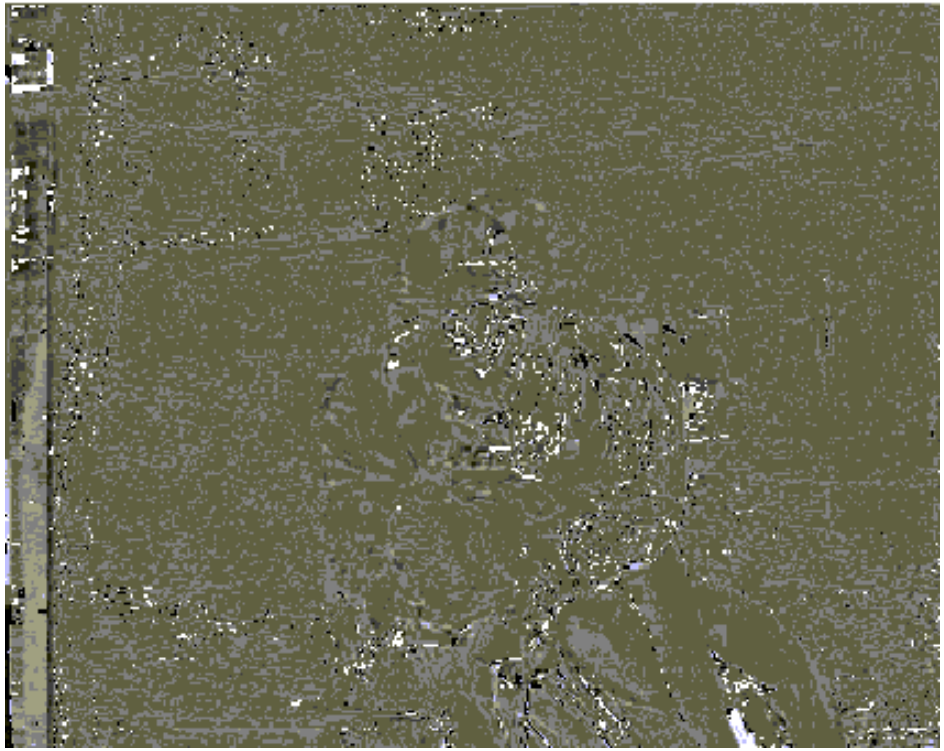


Frame N to be encoded

- Frame at t=N-1 used to predict content of frame N



Prediction error without motion compensation



Prediction error with motion compensation

MPEG 2 H.262

MPEG2

- MPEG-1 was suitable for storage media. Was aimed at VHS quality at 1.5 Mbps

- MPEG2 was designed as a superset of MPEG1 with support for storage media like DVD, set-top boxes and broadcast video at 4-9 Mbps, HDTV up to 60 Mbps, etc. Broadcast quality was obtained using fields instead of frames.

- MPEG2 supports higher bit rates and a larger number of applications:
  - Interlaced and progressive video
  - Different color sampling modes: 4:2:0, 4:2:2, 4:4:4
  - Predictive (P) and interpolative (B) coding
  - Flexible quantization schemes: can be changed at picture level
  - Scalable bit-streams
  - Profiles and levels

MPEG2 and future

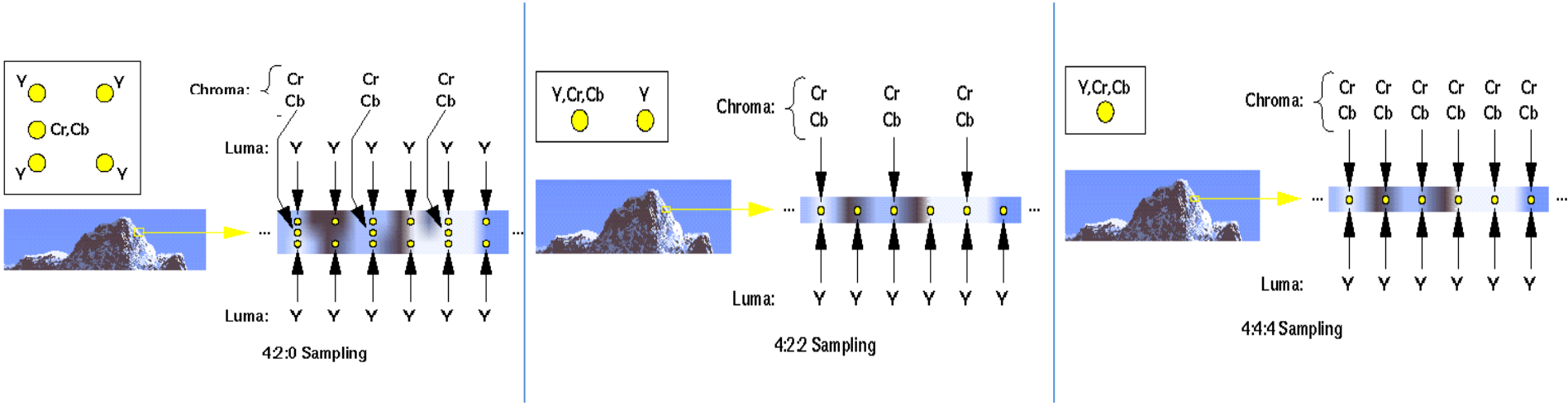
- MPEG2 H.262 coder is relatively simple and provides very high quality compression.

- However MPEG-2 video is not optimized for low bit-rates (less than 1 Mbit/s). Moreover the size of the compressed file is large and unsuited for media that are streamed on the internet.
  Due to this MPEG-4 H.264 is the standard for mobile devices and internet streaming

- On March 8, 2022 Digital Terrestrial TV has changed again in favor of MPEG-4 and will change again in 2023 to HEVC (High Efficiency Video Coding to allow to manage more effectively 5G

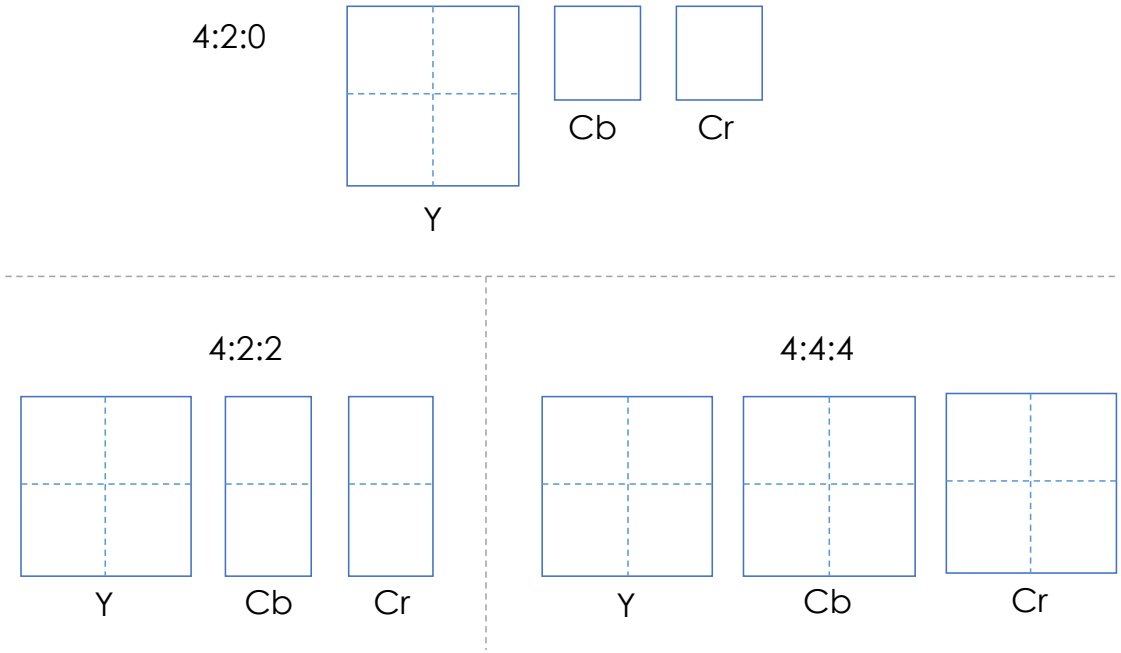
Color subsampling

- MPEG2 supports different color subsamplings:
  - 4:2:0
    with chrominance samples horizontally and vertically co-located on luminance samples
  - 4:2:2, 4:4:4
    Allow professional quality
    Different quantization matrices for Y and CrCb with 4:2:2 and 4:4:4 sampling



Macroblock structure

- 4:2:0, 4:2:2 and 4:4:4 sampling determine different mcroblock structures for Y, Cr and Cb



I, P, B frame encoding

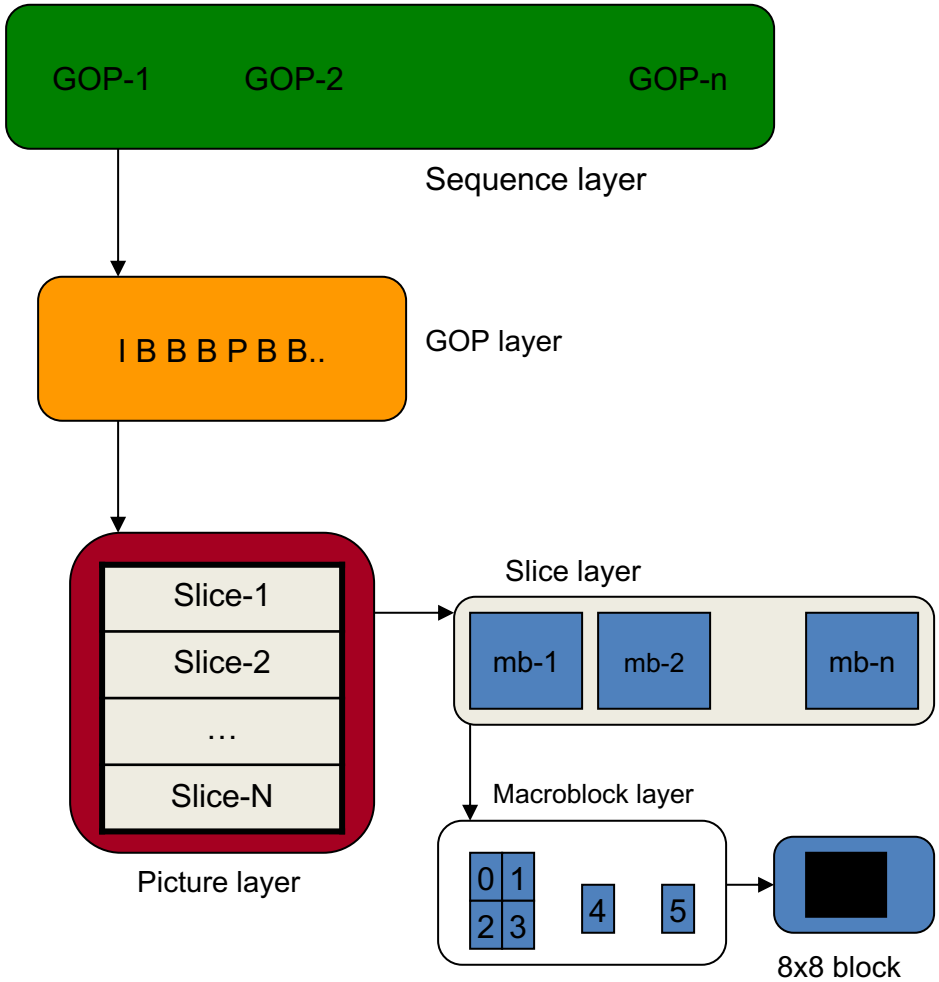
- Same as MPEG1. I, P and B frames (pictures) are encoded on a macroblock basis. There is no support of D pictures

- I-pictures have intraframe coding, using DCT
- P-pictures have interframe predictive coding
  - Macroblocks may be coded with forward prediction from references made from previous I and P pictures (P-type) or may be intra coded (I-type)
- B-pictures have interframe interpolative coding
  - Macroblocks may be:
    - P-type: coded with forward prediction from past I or P references;
    - B-Type: coded with interpolated prediction from past (forward) and future (backward) I or P references;
    - I-type: intra coded (no prediction).

- For each macroblock the motion estimator produces the best matching macroblock. The prediction error is encoded using a block-based DCT

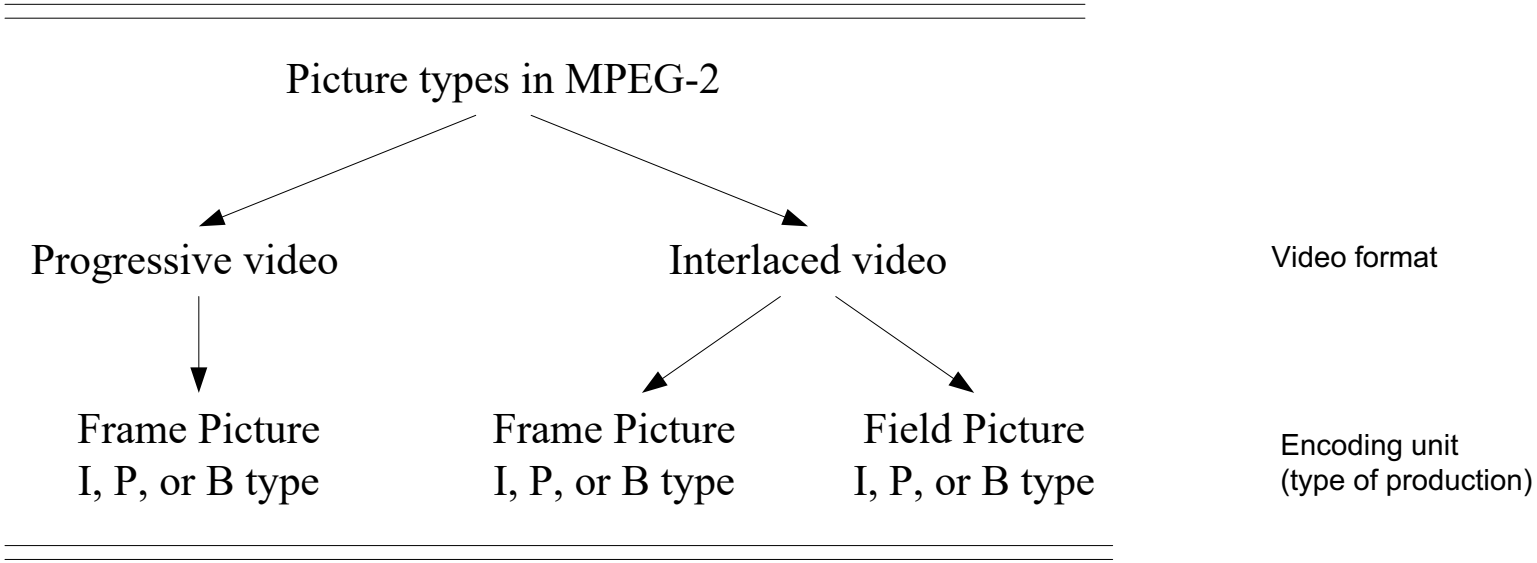
MPEG2: Structure of the bit-stream

- *Sequence layer*: picture dimensions, pixel aspect ratio, picture rate, minimum buffer size, DCT quantization matrices
- *GOP layer*: unit for video random access; the smallest unit of independent coding. Contains encoded pictures (type I, P, B)
- *Picture layer*: primary coding unit. [temporal frame ref number, picture type, synchronization info, resolution, ….]
- *Slice layer*: syncronizzation unit. [position of slice in picture, quantization scale factor]
- *Macroblock layer*: motion compensation unit. [position of macroblock in slice, motion vectors (H, V)]
- *Block layer* : unit for DCT processing



Progressive and interlaced video coding

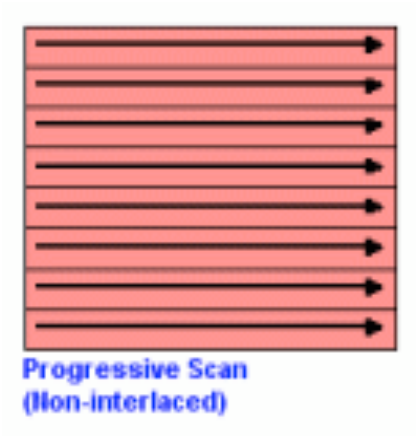
- MPEG2 supports both progressive and interlaced video.



Progressive video coding

- Progressive frames are encoded as *frame pictures* with frame-based macroblocks only. The 8x8 four blocks that compose the macroblock come from the same frame of video

  With frame production of progressive video only *frame-based macroblock prediction* is possible. *Frame-based prediction* uses a single motion vector for each 16x16 macroblock.

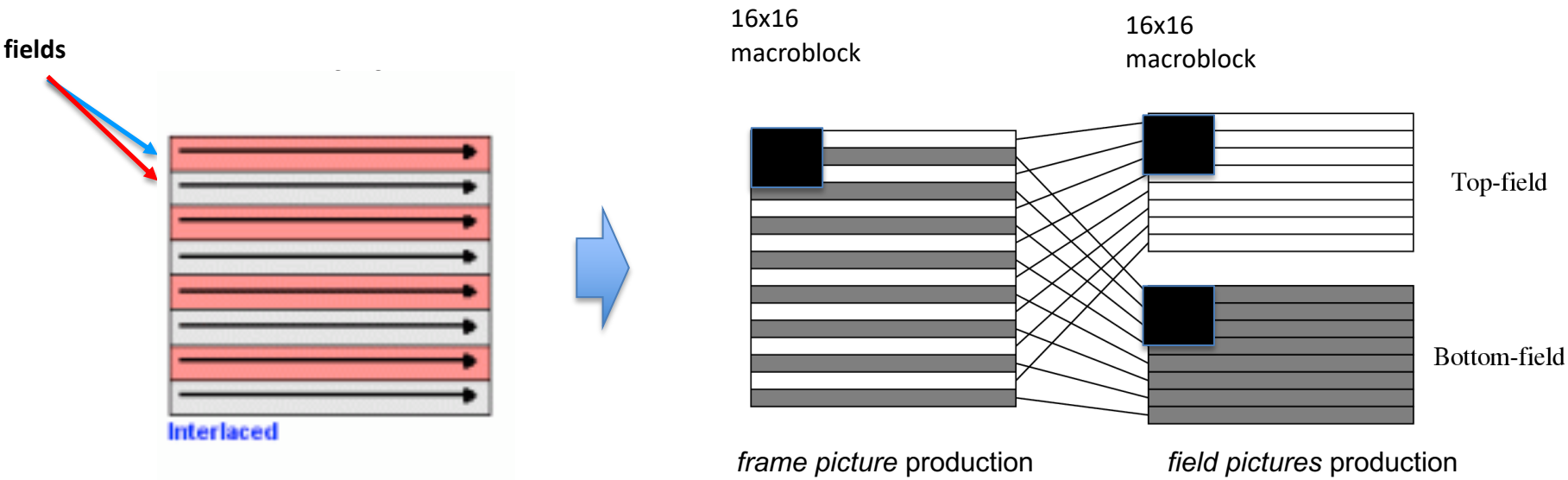


16x16
macroblock

frame picture production

Interlaced video coding

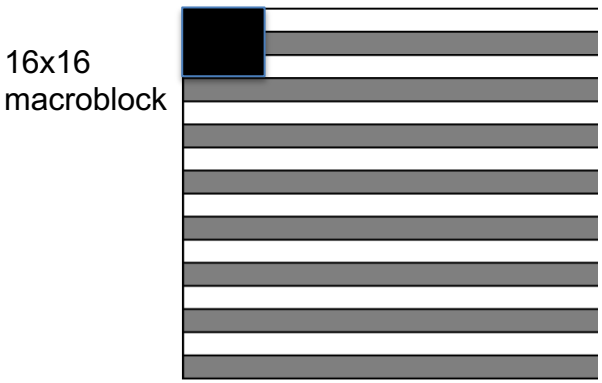
- Interlaced frames may be encoded as either a *frame picture* or as *two separately coded field pictures.* The encoder may decide on a frame by frame basis to produce a frame picture or two field pictures.
  - In the case of a *frame picture* is produced**,** all scanlines from both fields are interleaved to form a frame and the frame is divided into 16x16 macroblocks for coding.
    With frame production of interlaced video it is possible to use either *frame-based* or *field-based macroblock prediction*
  - In the case of *field pictures* are produced, the field is treated as a separate picture and is divided into 16x16 macroblocks for coding.
    With field production of interlaced video it is possible to use only *field-based macroblock prediction*



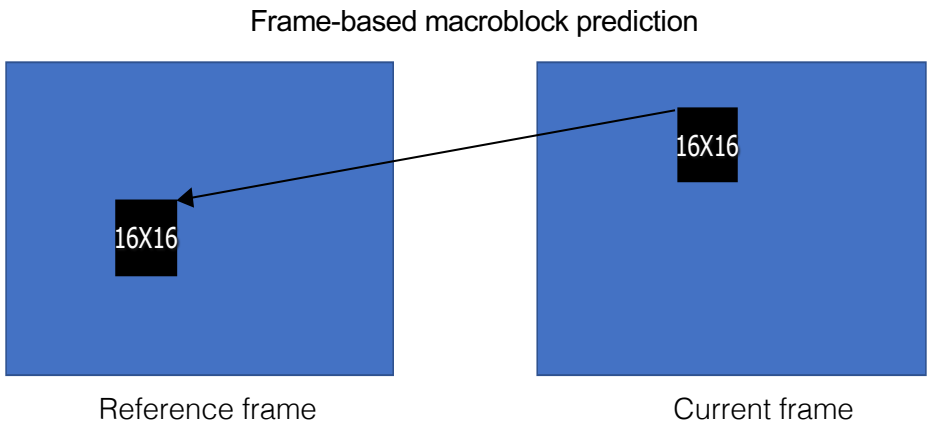
Frame production: frame-based prediction

progressive and interlaced video

- With *Frame production and frame-based macroblock prediction* a single motion vector is used for each 16x16 macroblock. Frame-based macroblock prediction is suited for macroblocks with little motion and high spatial activity.

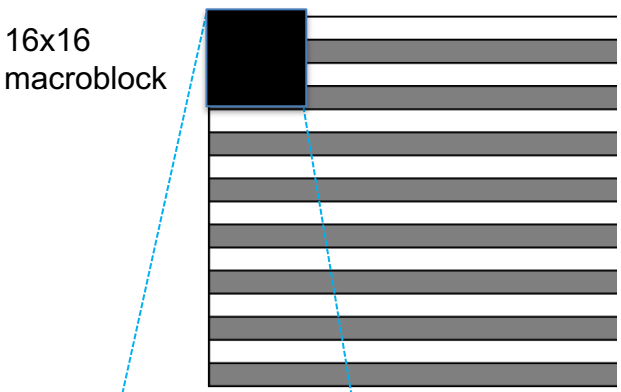


Frame picture production

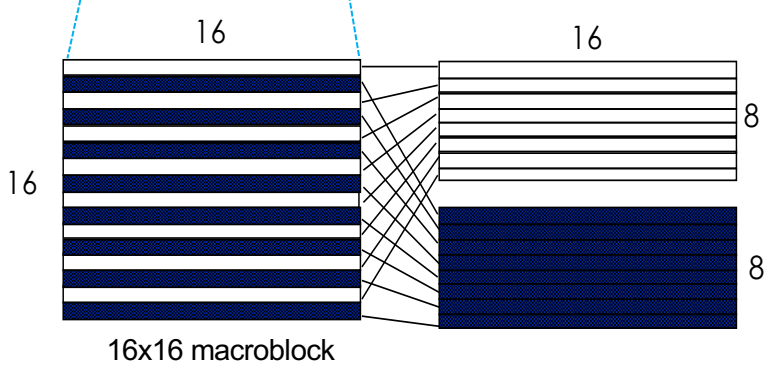


Frame production: field-based prediction
interlaced video

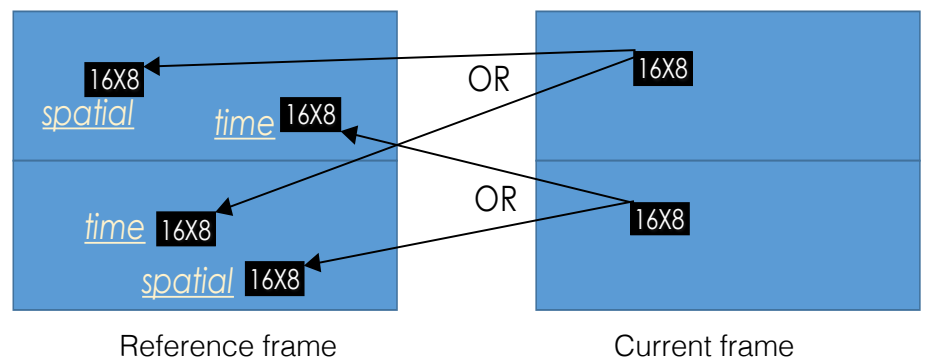
- With *Frame production* and *field-based macroblock prediction* the top and bottom field are treated separately. Field-based macroblock prediction is suited for high motion macroblocks.
- Each 16 × 16 macroblock is split into two 16 × 8 parts, each coming from one field. Two motion vectors are used: the first refers to the upper 16x8 region, the second to the lower 16x8 region. Each field is predicted separately with its motion vectors.



Frame picture production



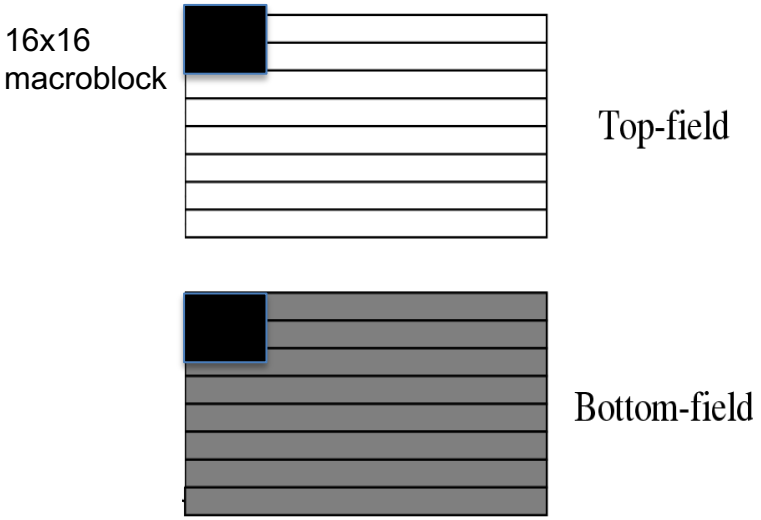
Field-based macroblock prediction



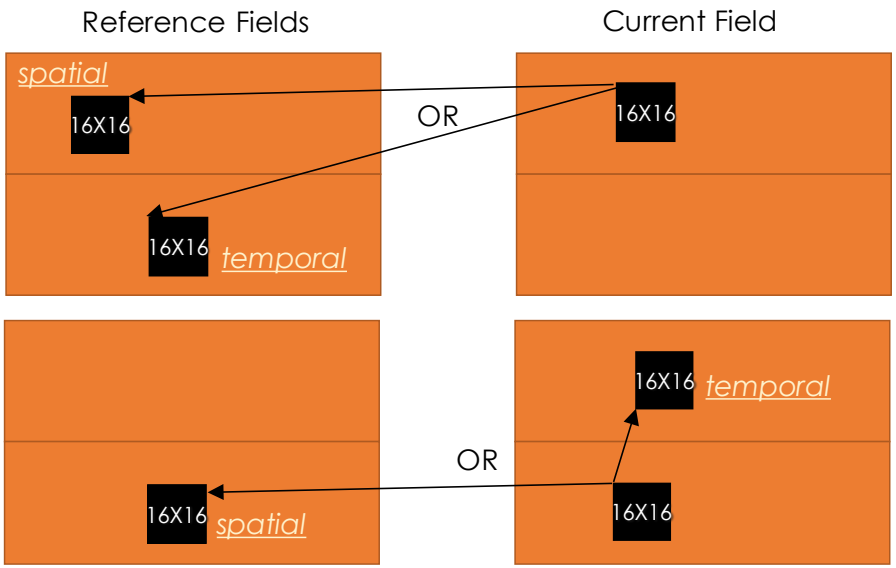
Field production: field-based prediction

interlaced video

- With *Field-production*, *field-based macroblock prediction* must be used based on a macroblock of size 16 × 16 from field-pictures.



Field pictures production

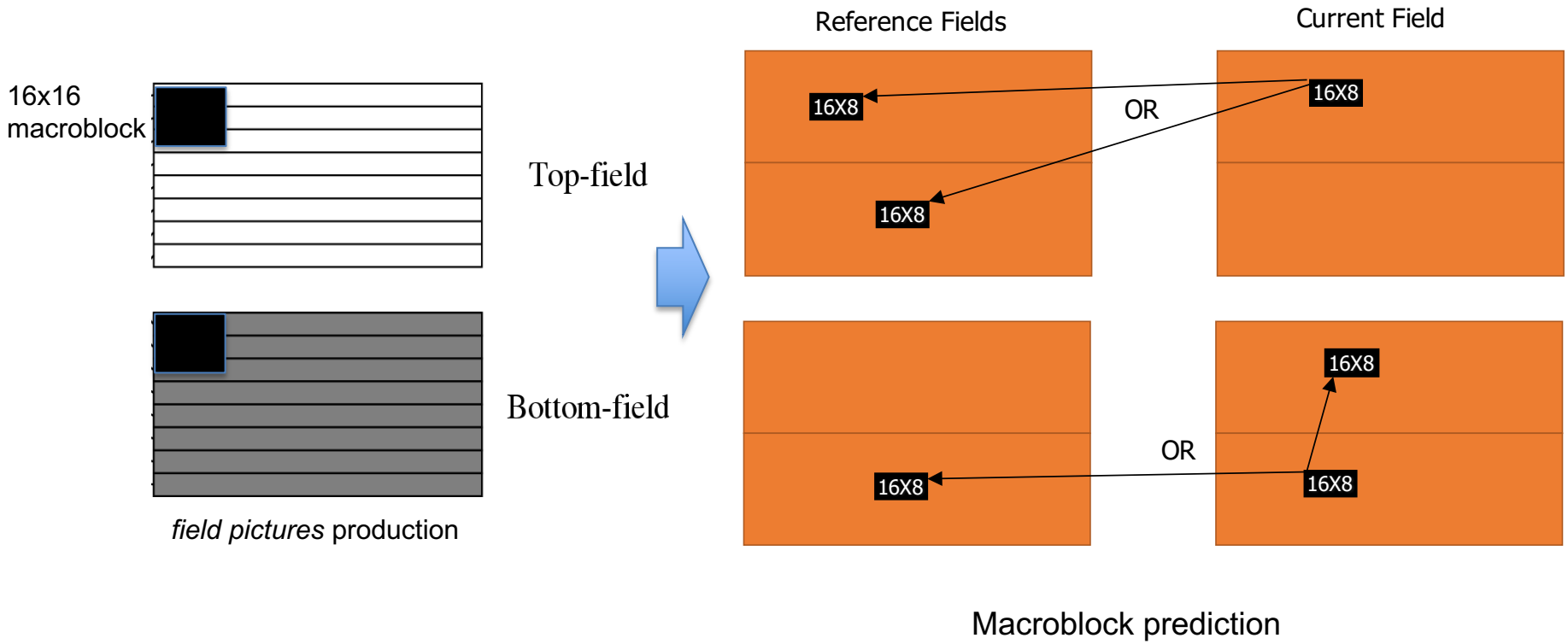


Field-based macroblock prediction

Field production: field-based prediction

interlaced video

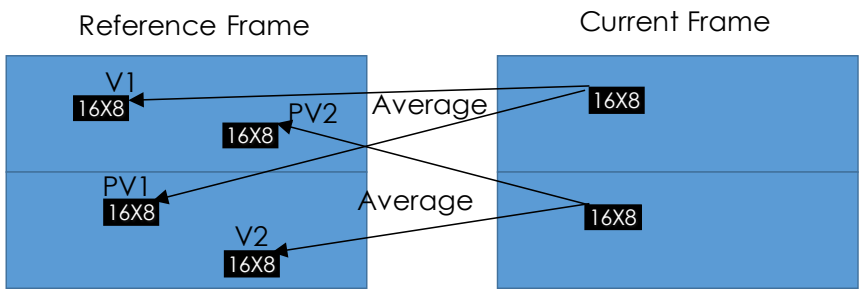
- The size of 16 × 16 in the field picture covers a size of 16 × 32 in the Frame picture. It is too big size to assume that behavior inside the block is homogeneous. 16 × 8 size prediction was introduced in field picture for the case of a lot of irregular motion



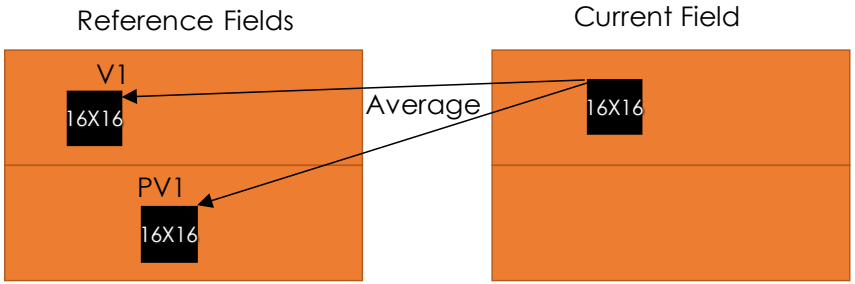
Dual-prime prediction
interlaced video

- *Dual-prime macroblock prediction* is a prediction mode in which two forward field-based predictions are averaged. So only one motion vector is transmitted per motion block for P-pictures. The predicted block size is 16x16 luminance samples.
- It is the only mode that can be used for either frame-pictures or field-pictures production.

- It is only used in interlaced P-pictures when there have been no B-pictures between the P-picture and its reference frame. It is used as an alternative to bi-directional prediction, where low delay is required. It avoids the frame re-ordering needed for bi-directional prediction but achieves similar coding efficiency.



With frame pictures production



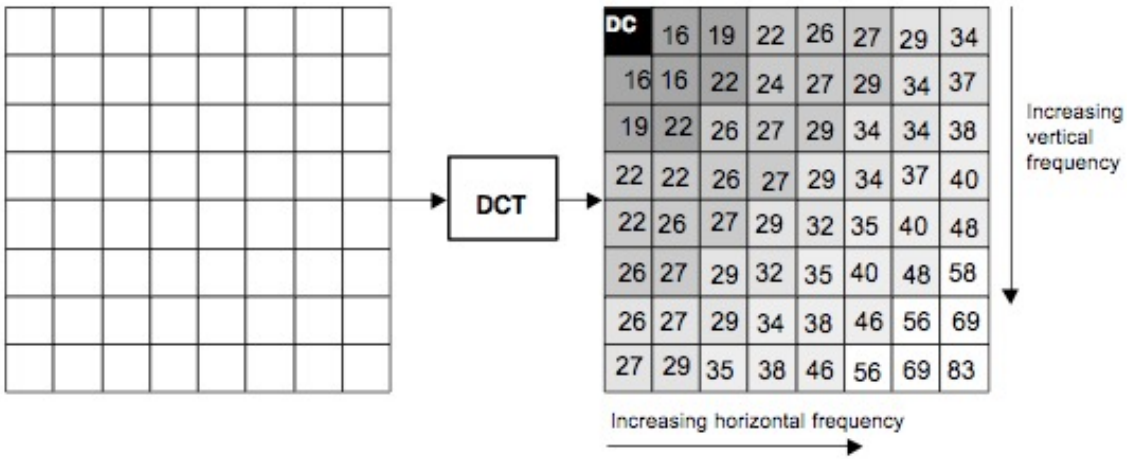
With field pictures production

Discrete Cosine Transform and quantization scale

8x8 block

Quantization matrix

It



Non linear quantization scale is also available

8 x 8 DCT:

$$F(u,v) = \frac{C(u)C(v)}{4}\left[\sum_{x=0}^{7}\sum_{y=0}^{7} f(x,y)\cos\left(\frac{(2x+1)u\pi}{16}\right)\cos\left(\frac{(2y+1)v\pi}{16}\right)\right], \qquad u,v = 0,\ldots,7$$

8 x 8 IDCT:

$$f(x,y) = \frac{1}{4}\left[\sum_{u=0}^{7}\sum_{v=0}^{7} C(u)C(v)F(u,v)\cos\left(\frac{(2x+1)u\pi}{16}\right)\cos\left(\frac{(2y+1)v\pi}{16}\right)\right], \qquad x,y = 0,\ldots,7$$

Multiple scanning options

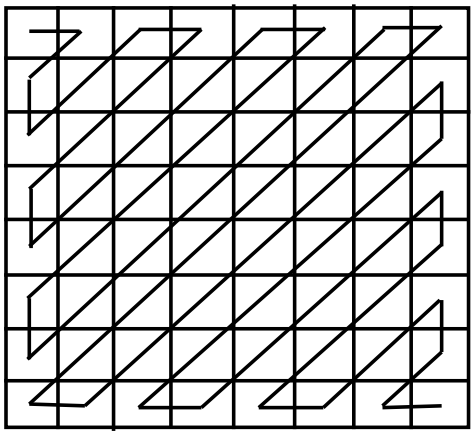
- zig-zag scanning is accompanied with a different scanning that is better suited for interlaced frames

-42	+12	+3	-1	0	0	0	0
-19	+14	-8	0	0	0	0	0
+16	+8	-9	+3	0	0	0	0
+3	-2	0	0	+1	0	0	0
+3	0	0	+1	-1	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

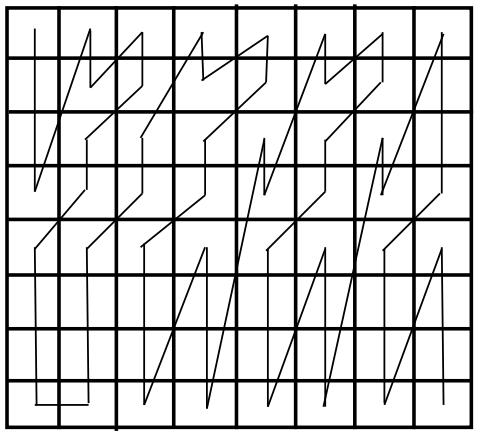
Increasing vertical frequency

Increasing horizontal frequency

Zigzag (progressive)



Alternate (interlaced)

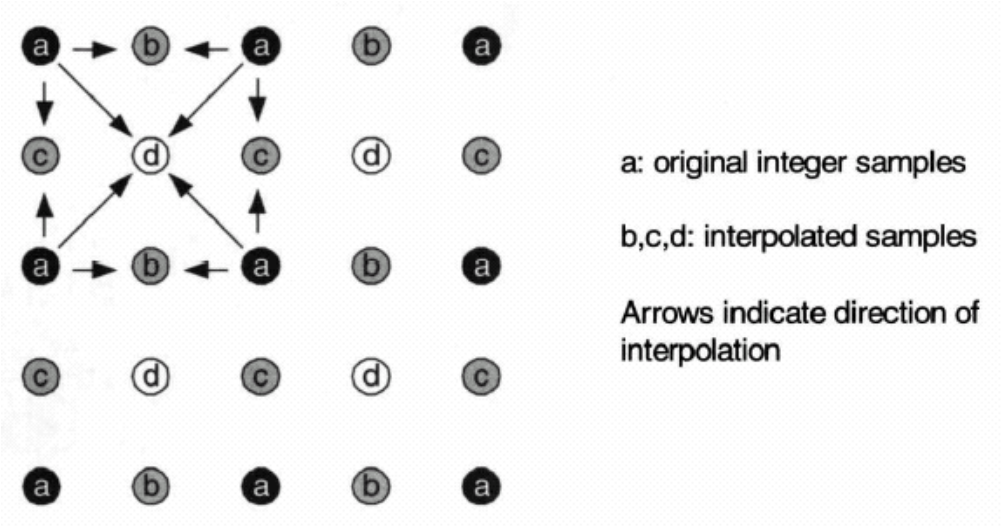


Half pixel interpolation for motion estimation

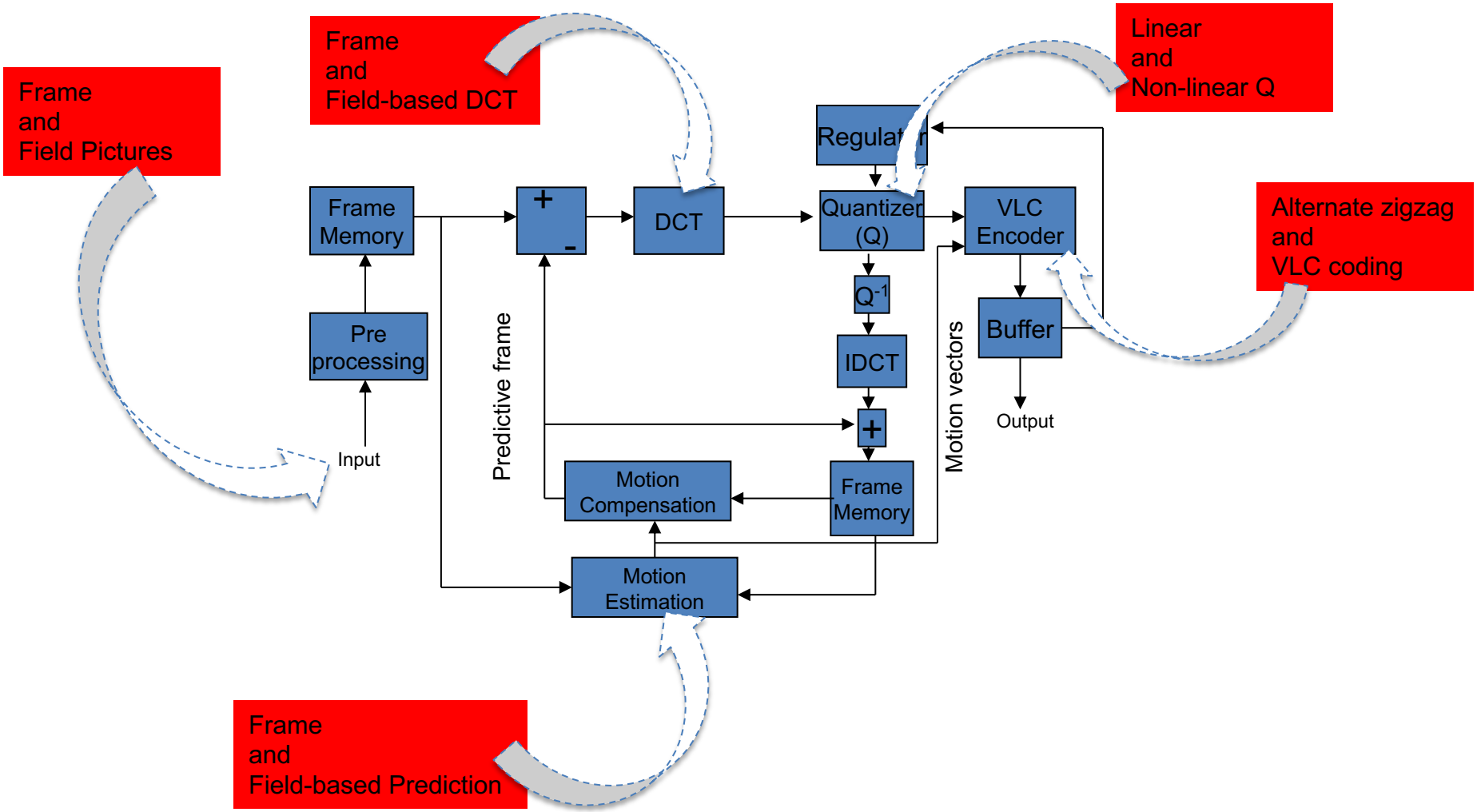
- MPEG2 uses half-pixel interpolation for motion vector estimation.

- Searching is performed as follows:
  - Pixels are interpolated in the image search area so that a region is created with higher resolution than the original.
  - Best match search is performed using both pixel and subpixel locations in the interpolated region
  - Samples of the best matched region are subtracted from the samples of the current block to obtain the error block.

Half pixel interpolation



MPEG2 Enhancements



Profiles and Levels

- In MPEG2 profiles and levels (profile@level) define the minimum capability required for the decoder:
  - Profiles: specify syntax and algorithms (define the compression rate and decoding complexity)
  - Levels: define parameters such as resolution, bitrate, etc.

- Simple Profile (4:2:0)
  - For videoconferencing
  - Corrisponds to MPEG1 Main profile without B frame
- Main profile (4:2:0)
  - For videoprofessional SDTV (bitrate at 50 Mbps)
  - The most important; of general applicability
- Multiview profile
  - For multiple cameras filming the same scene.
- 4:2:2 profile
  - For video professional SDTV and HDTV (bitrate at 50 Mbps)
- SNR and Spatial Scalable profile (4:2:0)
  - Add SNR / spatial scalability SNR with different quality levels
- High 4:2:0 profile
  - Suitable for HDTV

- Low Level
  - MPEG1 CPB (Constrained Parameters Bitstream): max. 352x288 @ 30 fps
- Main Level
  - MPEG2 CPB (720x576 @ 30 fps)
- High-1440 and High Levels
  - Typical of HDTV

Profiles and levels

Level	Profile				
	Simple 4:2:0	Main 4:2:0	SNR Scalable 4:2:0	Spatially Scalable 4:2:0	High 4:2:0 or 4:2:2
High 1920x1152 (60 frames/s)		80 Mbit/s			100 Mbit/s for 3 layers
High-1440 1440x1152 (60 frames/s)		60 Mbit/s		60 Mbit/s for 3 layers	80 Mbit/s for 3 layers
Main 720x576 (30 frames/s)	15 Mbit/s	15 Mbit/s	15 Mbit/s for 2 layers		20 Mbit/s for 3 layers
Low 352x288 (30 frames/s)		4 Mbit/s	4 Mbit/s for 2 layers		

* numbers in the table are maximum allowed

MPEG-2 scalability

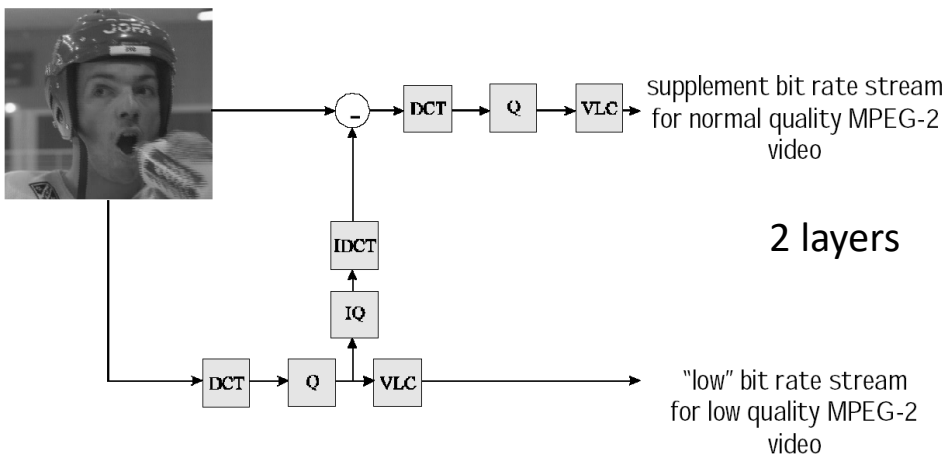
- Scalability is the ability of decoding only part of the stream to obtain a video of the resolution desired. It is possible to have:
  - SNR scalability,
  - Spatial scalability
  - Temporal scalability

- Scalability mode permits interoperability between different systems (f.e. a HDTV stream is also visible with SDTV). A system that does not reconstruct video at higher resolution (spatial or temporal) can simply ignore data refinement and take the base version.

SNR-scalability

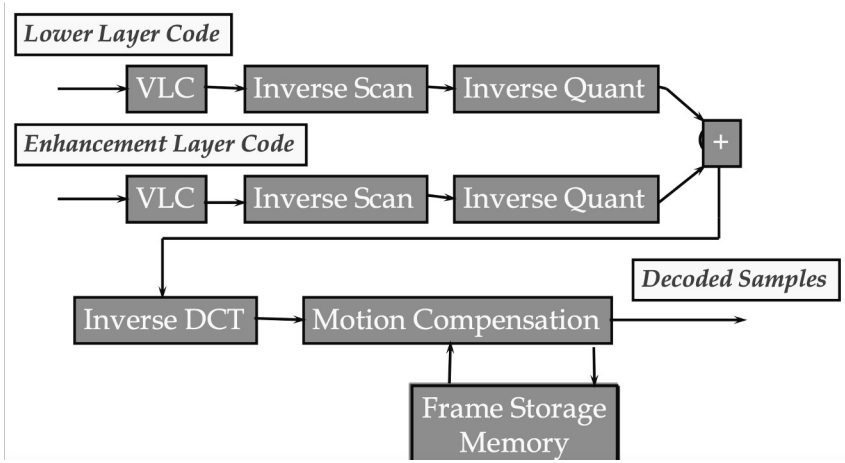
- Suited for applications that require different degrees of quality. Permits graceful degradation

- SNR Scalability encoding
  - All layers have the same spatial resolution.
  - The base layer provides the base quality, the enhancement layer provides quality improvements Refinement of the DCT coefficients encoded in base layer by the enhancement layer coefficients
- SNR Scalability Decoding
  - DCT coefficients in base layer are added to DCT coefficients in enhancement layer



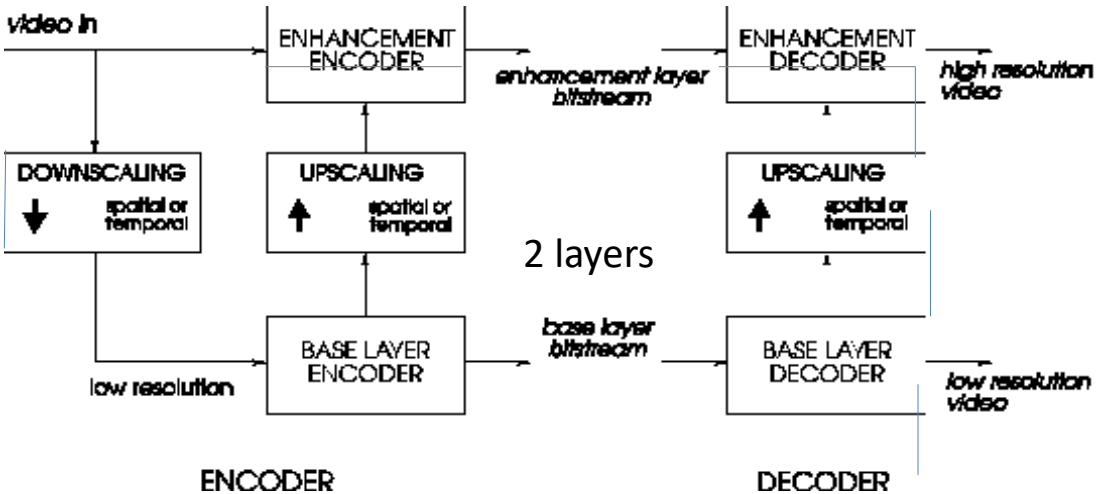
Encoder

Decoder



Spatial and temporal scalability

- Spatial scalability (2 layer)
  - Base layer at lower spatial resolution (MPEG1 can be used to encode the base layer)
  - Enhancement layer at higher resolution (obtained by spatial interpolation). Upscaling is used to predict coding of the high resolution version. Prediction error is encoded in the enhancement layer bitstream



- Temporal scalability (2 layer)
  - Similar to spatial scalability, but referred to time
  - Base Layer : 15 fps
  - Enhancement layer: supplements the remaining frames to achieve higher fps

MPEG2 criticals

- There are several conditions that are critical for MPEG2 compression:
  - Zooming
  - Rotations determine *mosquito noise*
  - Non-rigid motion

  - Dissolves and fades determines *blockness*

  - Shadows
  - Smokes
  - Scene cuts
  - Panning across crows determine *wavy noise*
  - Abrupt brightness changes
  - …….